

No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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1 Abstract

Stackelberg Security Games (SSGs) model the strategic allocation of limited defender resources against attackers whose preferences are often uncertain. Balcan et al. [2] designed no-regret online learning algorithms for the *repeated* variant of SSGs against an adversarial sequence of attackers drawn from a finite, *known* set K , and proved a matching linear-regret lower bound when K is unbounded. A natural real-world middle ground is left open: the defender does not know K a priori, but has a bound $K_{\max}$ on its size. We study this setting in the full-information feedback model. We propose GROW-CATALOGUE, an algorithm that incrementally learns new attacker types as they arrive, re-partitions the coverage polytope, and runs an anytime multiplicative-weights sub-routine on the resulting extreme-point set. Our central technical contribution is a full proof that GROW-CATALOGUE achieves expected regret

$$O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right)$$

against any oblivious adversarial sequence of length T . The rate matches that of Balcan et al. [2] with k replaced by $K_{\max}$ and is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$. Experiments on synthetic SSG instances confirm the predicted $\sqrt{T}$ scaling, and the empirical regret stays comfortably below the analytic $C\sqrt{T}$ envelope. Our implementation and all experiment scripts are at github.com/aditisingh1053/Security_Games.

2 Introduction

In a Stackelberg Security Game (SSG), a defender allocates a scarce set of resources to protect n potential targets against a strategic attacker who, observing the defender's randomized deployment, chooses the best target to attack [19]. Deployed systems built around this model include the US Coast Guard's port protection patrols, the Federal Air Marshals assignment system, the LAX airport patrol schedule, and anti-poaching patrols in several African national parks [19, 20]. In every one of these applications the defender must commit to its deployment before the attack happens, and the attacker's payoff function encodes its preferences over targets.

The *classical* SSG analysis assumes that the attacker's payoff function is fully known to the defender. The optimal commitment can then be computed by a linear program [e.g., 18]. In practice, however, the defender faces considerable uncertainty about the attacker. There may be several distinct threat actors (e.g., different poacher groups, different types of smugglers), and their utilities are only imperfectly known. A principled response, pioneered by Balcan et al. [2], is to drop the single-type assumption entirely and instead ask for

a *no-regret* online algorithm against an adversarially chosen sequence of attacker types, drawn from a known finite set K with $|K| = k$. Under full-information feedback, they design an algorithm whose expected cumulative regret is bounded by $O(\sqrt{Tn^2k \log(nk)})$, which is sublinear in the horizon T , and they prove a matching linear-regret lower bound when $|K|$ is unrestricted [2, Theorem 7.1].

The assumption that the set K itself is known to the defender in advance is, however, a strong one. In realistic deployments the defender typically does *not* have a complete list of attacker types at the start. What it may have is an *upper bound* $K_{\max}$ on the number of distinct types it expects to encounter (for instance, from threat intelligence or from physical constraints of the domain). The question that motivates this project is therefore:

Can the defender achieve sublinear regret when K is unknown a priori, subject to $|K| \leq K_{\max}$?

This question lies squarely in the gap left open between the *known- K* upper bound and the *unrestricted- K* lower bound of Balcan et al. [2]. It is not answered by any of the existing work, and resolving it is the main goal of the present report.

2.1 Contributions

Our contributions are the following.

- **Problem formalization (Section 4).** We formalize the unknown- K variant of the repeated Stackelberg security game, identifying the minimal modification of the model of Balcan et al. [2] that captures it.
- **Algorithm (Section 5).** We propose GROW-CATALOGUE, an online algorithm that at each round maintains a *catalogue* $C_t \subseteq K$ of attacker types observed so far, recomputes the set $\mathcal{E}(C_t; \epsilon)$ of ϵ -approximate extreme points of the partition of $\mathcal{P}$ induced by C_t , and runs an anytime multiplicative-weights sub-routine over $\mathcal{E}(C_t)$. When a new type is observed, the catalogue is updated and the sub-routine is restarted on the enlarged expert set.
- **Main theorem and a complete, detailed proof (Sections 5.2 and 6).** The technical core of this report is a self-contained proof that GROW-CATALOGUE achieves expected regret at most $O(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)})$ against any oblivious adversarial sequence of length T under full-information feedback. The rate matches Theorem 5.1 of Balcan et al. [2] with k replaced by $K_{\max}$; for constant $K_{\max}$ it is $\tilde{O}(n\sqrt{T})$.
- **Regime analysis (Corollaries 7 and 8).** We characterize the parameter regime in which the bound is meaningful ($K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$), and we prove an instance-adaptive refinement in which the leading factor depends on

the number K' of types *actually* observed rather than on $K_{\max}$.

- **Experiments (Section 7).** We implement GROW-CATALOGUE and two reference baselines, and evaluate them on synthetic SSG instances. The empirical regret scales as predicted by the theorem, staying comfortably below the analytic $C\sqrt{T}$ envelope.

2.2 Related Work

Classical SSGs and attacker uncertainty. The optimal commitment for a known attacker can be computed by an LP [5]. Uncertainty about the attacker has been handled by *robust optimization*, which plans against worst-case payoffs [18], and by *Bayesian SSGs*, which require a prior over attacker types [12, 15]. Both approaches rest on priors or constraints that are often hard to validate. Letchford et al. [13] give a query-efficient learning algorithm for a *single* fixed but unknown type, and Blum et al. [3] extend this to a setting that adapts round-by-round to one fixed type.

Online learning against adversarial attacker sequences. Balcan et al. [2] pivot from the single-type paradigm by allowing the sequence of attackers to be adversarially chosen from a known finite set K . For full information they give a polynomial-weights algorithm with regret $O(\sqrt{Tn^2k \log(nk)})$ that partitions the coverage polytope into best-response regions and runs experts over the ϵ -approximate extreme points; for bandit feedback they use a barycentric spanner construction to give an $O(T^{2/3})$ bound. Crucially, they also prove a linear lower bound when $|K|$ is unrestricted, which forces any non-trivial extension to restrict the attacker type set somehow. We take the weakest restriction: a known cardinality bound $K_{\max}$.

Online learning with changing expert sets. The subroutine inside GROW-CATALOGUE is an anytime variant of Polynomial Weights / Hedge [4, 6, 14]. We restart this subroutine each time the catalogue grows; simpler algorithms for tracking a growing set of experts exist [16], and for tracking a shrinking set [7], but both would only shave lower-order logarithmic factors.

Other recent directions. Haghtalab et al. [8] analyze no-regret learning against strategic (non-myopic) attackers in a single-type query model. Harris et al. [9] extend to a contextual Stackelberg setting in which both players observe side information. Neither addresses the *unknown K with bounded cardinality* question that we pose here.

3 Model

We adopt the repeated-SSG notation of Balcan et al. [2, Sec. 2]; our notation is compatible with theirs.

3.1 Game primitives

The game has a finite horizon $T \in \mathbb{N}$ and n targets $N = \{1, \dots, n\}$. The defender has R resources and a collection $\mathcal{D} \subseteq 2^N$ of *schedules*, where each schedule is a subset of

targets that a single resource can defend simultaneously. An *assignment function* $A : R \rightarrow 2^{\mathcal{D}}$ records the schedules that each resource may be assigned to. A *pure strategy* is any valid resource-to-schedule assignment; let m denote the number of pure strategies. A *mixed strategy* is a distribution $\mathbf{s} \in \Delta_m$ over pure strategies, and its induced *coverage probability vector* is $\mathbf{p} = M\mathbf{s} \in [0, 1]^n$, where $M \in \{0, 1\}^{n \times m}$ has $M_{ij} = 1$ iff the j -th pure strategy covers target i . Write $\mathcal{P}$ for the set of valid coverage probability vectors; Every coverage vector $\mathbf{p}$ is realizable by some mixed strategy, and conversely every mixed strategy induces a unique $\mathbf{p} \in \mathcal{P}$.

Defender payoffs. For each target i , the defender has a payoff $u_d^c(i) \in [-1, 1]$ if i is attacked *and* covered, and a payoff $u_d^u(i) \in [-1, 1]$ if i is attacked *and* uncovered. The defender's expected utility when target i is attacked under coverage $\mathbf{p}$ is therefore

$$U_d(i, \mathbf{p}) = u_d^c(i) p_i + u_d^u(i) (1 - p_i), \quad (1)$$

which is *linear* in $\mathbf{p}$ for each fixed target i .

3.2 Attacker types and best response

An attacker *type* α is a pair $(u_\alpha^c, u_\alpha^u) \in [-1, 1]^n \times [-1, 1]^n$ of per-target utilities following the same “covered/uncovered” convention as the defender. Its expected utility for attacking target i under coverage $\mathbf{p}$ is $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i) p_i + u_\alpha^u(i) (1 - p_i)$. Observing $\mathbf{p}$, type α *best-responds*:

$$b_\alpha(\mathbf{p}) = \arg \max_{i \in N} U_\alpha(i, \mathbf{p}), \quad (2)$$

with ties broken arbitrarily but consistently. Note that b_α is piecewise constant in $\mathbf{p}$, which makes the defender's realized payoff $U_d(b_\alpha(\mathbf{p}), \mathbf{p})$ a discontinuous function of $\mathbf{p}$. This non-continuity is the central technical difficulty that distinguishes SSGs from vanilla online linear optimization, and it is what drives the polytope-based machinery of Balcan et al. [2].

3.3 Repeated game and regret

There is a (possibly unknown) set K of attacker types. An *oblivious* adversary chooses a sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ before the game starts; the sequence is fixed but unknown to the defender. At each round $t \in \{1, \dots, T\}$ the defender commits $\mathbf{p}_t \in \mathcal{P}$, attacker a_t plays $b_{a_t}(\mathbf{p}_t)$, and the defender receives the random payoff $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$. We assume *full-information feedback*: at the end of round t the defender observes the identity of a_t , and the first time any type α is observed, its utility functions are revealed as well.

The defender's benchmark is the best fixed coverage vector in hindsight:

$$\mathbf{p}^* = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}), \mathbf{p}). \quad (3)$$

The expected regret of an online algorithm that plays $\mathbf{p}_t$ at round t is

$$\text{Regret}(T) = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right], \quad (4)$$

where the expectation is over the algorithm's internal randomization.

3.4 Preliminaries: polytope partition and $\mathcal{E}(C; \epsilon)$

The following definitions, taken from Balcan et al. [2, Sec. 4], are essential. For any catalogue $C \subseteq K$, a type $\alpha \in C$, and a target $i \in N$, the region in which α 's best response is i is

$$\mathcal{P}_i^\alpha(C) = \{\mathbf{p} \in \mathcal{P} : b_\alpha(\mathbf{p}) = i\}, \quad (5)$$

which is the intersection of $\mathcal{P}$ with the $n-1$ linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$ for $j \neq i$; hence a convex polytope. For a response profile $\sigma : C \rightarrow N$ we write

$$\mathcal{P}_\sigma(C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(C). \quad (6)$$

The non-empty sets $\mathcal{P}_\sigma(C)$ form a partition of $\mathcal{P}$ into convex regions in which the best response of every type in C is held fixed; within any such region, the defender's payoff under a fixed attacker type is a linear function of $\mathbf{p}$.

We will need three results from Balcan et al. [2], restated below in our notation and cited explicitly at every use. Lemmas 1 and 2 describe the approximate extreme-point set; Proposition 3 is the standard Polynomial-Weights online learning guarantee, adapted to the present loss range.

Lemma 1 (2, Lemma 4.3). *For every $\epsilon > 0$ and every catalogue $C \subseteq K$, there is a finite set $\mathcal{E}(C; \epsilon) \subseteq \mathcal{P}$ of points, called the ϵ -approximate extreme points of the partition, such that for any sequence $\mathbf{a}' = a'_1, \dots, a'_{T'}$ with $a'_t \in C$ for all t ,*

$$\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T'.$$

Lemma 2 (2, Lemma 4.4). *The set $\mathcal{E}(C; \epsilon)$ of Lemma 1 can be chosen so that $|\mathcal{E}(C; \epsilon)| \in O((2^n + |C|n^2)^n \cdot n^{|C|})$.*

Proposition 3 (2, Proposition 1). *For any fixed expert set $\mathcal{M}$ and any sequence of losses in $[-\kappa, \kappa]$, there is an online learning algorithm (an anytime variant of Polynomial Weights / Hedge; see 4, Sec. 2) whose expected regret after T' rounds is at most $O(\sqrt{T' \kappa \log |\mathcal{M}|})$, without requiring T' to be known in advance.*

4 The Unknown- K Setting

We now state the problem that our algorithm and theorem address.

Problem 4 (Unknown- K repeated SSG). A repeated SSG as in Section 3 is played, with two modifications.

1. The set K of attacker types is not revealed to the defender at the start; however, the defender is given a constant $K_{\max} \geq 1$ with the promise $|K| \leq K_{\max}$.
2. Feedback is full information: after round t the defender observes a_t , and the first time any type α appears, its utility vectors (u_α^c, u_α^u) are revealed.

The benchmark $\mathbf{p}^*$ and regret $\text{Regret}(T)$ are as in Equations (3) and (4).

Algorithm 1 GROW-CATALOGUE (full information).

Require: horizon T , bound $K_{\max}$, game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

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1:  $C \leftarrow \emptyset$ ;  $\mathcal{E} \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/\sqrt{T}$ ;  $t_{\text{loc}} \leftarrow 0$ 
2: for  $t = 1, \dots, T$  do
3:   if  $\mathcal{E} = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     sample  $\mathbf{p}_t \sim q_t$ , where  $q_t$  is the current PW distribution
       over  $\mathcal{E}$ 
7:   end if
8:   observe  $a_t$ 
9:   if  $a_t \notin C$  then ▷ discovery round: new attacker type
10:     $C \leftarrow C \cup \{a_t\}$ ; record  $(u_{a_t}^c, u_{a_t}^u)$ 
11:     $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$  ▷ recompute extreme points
12:    reset PW on  $\mathcal{E}$ ;  $t_{\text{loc}} \leftarrow 0$ 
13:   else ▷ routine round
14:     $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$ 
15:    compute  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$  for  $\mathbf{p} \in \mathcal{E}$ 
16:     $\eta \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$ 
17:     $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} \cdot e^{-\eta \ell_t(\mathbf{p})}$  for all  $\mathbf{p} \in \mathcal{E}$ 
18:   end if
19: end for
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Remark 5 (Necessity of the bound $K_{\max}$). Balcan et al. [2, Theorem 7.1] exhibit, for every T , a game with $|K| < 2^{T+1}$ on which every online algorithm has expected regret at least $T/2$. Hence some finite bound on $|K|$ is *necessary* for sublinear regret; Problem 4 takes the weakest such bound, namely a known constant $K_{\max}$.

5 Algorithm and Main Result

5.1 The Grow-Catalogue algorithm

We describe GROW-CATALOGUE at a high level and then give the pseudocode in Algorithm 1. At any round t the defender holds a catalogue $C_t \subseteq K$ of all types seen so far. Conceptually, the defender runs a small instance of the *known- K* algorithm of Balcan et al. [2, Sec. 5] on the catalogue C_t , pretending that C_t is the full attacker set. Concretely:

- The algorithm maintains the ϵ -approximate extreme-point set $\mathcal{E}_t = \mathcal{E}(C_t; \epsilon)$ of Lemma 1. We set $\epsilon = 1/\sqrt{T}$; this balances the $2\epsilon T$ approximation term in Lemma 1 against the $\sqrt{T \log N_{\max}}$ regret term of Proposition 3.
- An anytime Polynomial-Weights sub-routine of Proposition 3 runs over $\mathcal{E}_t$, updating via a time-varying learning rate $\eta_{t_{\text{loc}}} = \sqrt{\log |\mathcal{E}_t| / t_{\text{loc}}}$ where t_{loc} is a local counter reset at every catalogue change. This is the standard anytime schedule; see Cesa-Bianchi and Lugosi [4, Sec. 2].
- When a new attacker type α appears, the catalogue grows to $C_t \cup \{\alpha\}$, the approximate extreme-point set is recomputed, and the sub-routine is restarted on the enlarged expert set. The single round of first contact with α contributes a bounded additive amount to the regret that we will absorb.

5.2 Main Result

Theorem 6 (Main). *Let the defender play Algorithm 1 in the unknown- K full-information setting of Problem 4. For any oblivious adversarial sequence $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$, the*

expected regret satisfies

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right). \quad (7)$$

In particular, if $K_{\max} = O(1)$ then $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(n\sqrt{T})$, matching the known- K bound of Balcan et al. [2, Theorem 5.1] up to logarithmic factors.

We prove Theorem 6 in detail in Section 6. Two useful corollaries of the proof (the regime of applicability and an instance-adaptive refinement) are stated after the proof in Section 6.1.

6 Proof of Theorem 6

The proof closely tracks the structure of the known- K proof of Balcan et al. [2, Sec. 5] but has two non-trivial deviations: (i) the expert set is no longer static but grows over time, and (ii) the discovery rounds introduce an extra bounded additive term. We account for both the deviations in our proof.

Proof:- Fix an oblivious adversarial sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ with $|K| \leq K_{\max}$. Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ denote the number of distinct types that actually appear in $\mathbf{a}$. The proof proceeds in ten steps.

Step 1 (Epoch decomposition). Let $\tau_1 < \tau_2 < \dots < \tau_{K'}$ be the round indices at which each new attacker type is first observed, i.e., for each $i \in \{1, \dots, K'\}$, round τ_i is the smallest t such that $a_t \notin \{a_1, \dots, a_{t-1}\}$. We define the *epochs*

$$\text{Ep}_i = \{\tau_i + 1, \tau_i + 2, \dots, \tau_{i+1} - 1\}, \quad i = 1, \dots, K', \quad (8)$$

with the convention $\tau_{K'+1} := T + 1$, and write $L_i = |\text{Ep}_i|$. Let $\mathcal{D} = \{\tau_1, \dots, \tau_{K'}\}$ be the set of *discovery rounds*. By construction,

$$[1, T] = \mathcal{D} \sqcup \bigcup_{i=1}^{K'} \text{Ep}_i,$$

and $|\mathcal{D}| + \sum_{i=1}^{K'} L_i = T$ with $|\mathcal{D}| = K' \leq K_{\max}$. The epoch decomposition is the critical structural property of the proof: it re-interprets the online problem with a growing expert set as a finite sequence of online problems with *fixed* expert sets, so that the known- K analysis of Balcan et al. [2] can be invoked inside each epoch.

Step 2 (Catalogue is constant inside an epoch). Immediately after round τ_i , the catalogue maintained by Algorithm 1 is $C_i = \{a_{\tau_1}, \dots, a_{\tau_i}\}$, and it is not updated until the next discovery round τ_{i+1} . Hence $C_t = C_i$ for every $t \in \text{Ep}_i$. Moreover, by the defining property of τ_{i+1} , no new type appears in Ep_i , so

$$a_t \in C_i \quad \text{for every } t \in \text{Ep}_i. \quad (9)$$

Denote by $\mathcal{E}_i := \mathcal{E}(C_i; \epsilon)$ the extreme-point set used by the PW sub-routine during Ep_i ; Equation (9) is exactly the hypothesis needed to apply Lemma 1 to C_i and the sub-sequence inside Ep_i .

Step 3 (Within-epoch regret). During Ep_i the algorithm runs a fresh instance of the anytime Polynomial-Weights sub-routine on the fixed expert set $\mathcal{E}_i$ for L_i rounds. The losses are $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so the range parameter is $\kappa = 1$. Proposition 3 therefore yields an absolute constant $c_1 > 0$ such that

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & - \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (10)$$

Step 4 (Within-epoch approximation via Lemma 1).

We now wish to compare the max over $\mathcal{E}_i$ during Ep_i with the *global* benchmark $\mathbf{p}^*$ (the best fixed strategy over the full horizon $[1, T]$). Lemma 1, however, is a statement of the form “ $\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)}(\cdot) \geq \max_{\mathbf{p} \in \mathcal{P}}(\cdot) - 2\epsilon T'$ ” and, strictly speaking, does *not* directly involve $\mathbf{p}^*$: it compares the max over the finite set $\mathcal{E}(C; \epsilon)$ against the unrestricted max over $\mathcal{P}$, both on the *same* attacker sequence. We therefore proceed in two sub-steps and make the passage to $\mathbf{p}^*$ explicit.

Sub-step 4a (apply Lemma 1 to the epoch sub-sequence). By Equation (9), the sub-sequence $(a_t)_{t \in \text{Ep}_i}$ has length L_i and uses only types in C_i , which is exactly the hypothesis of Lemma 1. Applying the lemma with catalogue C_i and sequence length $T' = L_i$ gives

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \underbrace{\max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p})}_{=: \text{OPT}_i} - 2\epsilon L_i. \end{aligned} \quad (11)$$

Here OPT_i is the best cumulative utility achievable on Ep_i by *any* coverage vector in $\mathcal{P}$; crucially, this is the epoch- i optimum, which in general *differs* from the value of $\mathbf{p}^*$. This is the subtle point that distinguishes our setting from the known- K proof of Balcan et al. [2, Sec. 5]: in their analysis, Lemma 1 is applied once on the full horizon, so $\text{OPT} = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*)$ by definition of $\mathbf{p}^*$ and no separate passage is needed. In our per-epoch application, OPT_i is an epoch-local quantity, potentially larger than $\mathbf{p}^*$'s value on the epoch.

Sub-step 4b (pass from OPT_i to $\mathbf{p}^$).* Since $\mathbf{p}^* \in \mathcal{P}$, the coverage vector $\mathbf{p}^*$ is a *feasible* competitor in the maximization defining OPT_i . Therefore

$$\text{OPT}_i \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (12)$$

Inequality Equation (12) is *not* a consequence of Lemma 1 but of the trivial fact that the global benchmark $\mathbf{p}^*$ is an element of the set over which the epoch-local optimum is taken. Chaining Equation (11) with Equation (12),

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i. \end{aligned} \quad (13)$$

Step 5 (Per-epoch combined bound). Chaining Equations (10) and (13), for every epoch $i \in \{1, \dots, K'\}$ we obtain

$$\begin{aligned} & \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (14)$$

Equation (14) is the per-epoch regret bound *against the global benchmark* $\mathbf{p}^*$, and it is the building block for the rest of the proof. Note that despite the fact that $\mathbf{p}^*$ is the benchmark over all T rounds (not just epoch i), we are able to bound our per-epoch regret against $\mathbf{p}^*$ because $\mathbf{p}^* \in \mathcal{P}$ and we applied Lemma 1 on the sub-sequence.

Step 6 (Discovery rounds). For each discovery round $t \in \mathcal{D}$, the algorithm either plays $\mathbf{p}_{\text{def}}$ (if $t = \tau_1$) or samples $\mathbf{p}_t$ from the PW distribution inherited from Ep_{i-1} (if $t = \tau_i$ for $i \geq 2$). In both cases the sampled $\mathbf{p}_t$ is not tuned to the newly-appearing type $a_t = a_{\tau_i}$. However, because the payoff U_d is bounded in $[-1, 1]$, for every $\mathbf{p} \in \mathcal{P}$ and every type α , $U_d(b_\alpha(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so

$$|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)| \leq 2. \quad (15)$$

Summing Equation (15) over the at most $K' \leq K_{\max}$ discovery rounds yields

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq 2K_{\max}. \quad (16)$$

This is the only source of additive dependence on $K_{\max}$ that does not come from the main $\sqrt{TK_{\max}}$ term. It is the “price of not knowing K ” at a discovery round.

Step 7 (Uniform bound on $|\mathcal{E}_i|$). By Lemma 2 applied with $|C_i| \leq K_{\max}$,

$$|\mathcal{E}_i| \leq c_2 (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} =: N_{\max},$$

for an absolute constant c_2 . Taking logarithms,

$$\begin{aligned} \log N_{\max} &= n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\leq n^2 + n \log\left(1 + \frac{K_{\max} n^2}{2^{n-1}}\right) + K_{\max} \log n \\ &= O(n^2 + K_{\max} \log n), \end{aligned} \quad (17)$$

where the middle step uses $\log(2^n + K_{\max} n^2) \leq n + \log(1 + K_{\max} n^2 / 2^{n-1})$ and $\log(1 + x) \leq x$ for $x > 0$. Crucially, Equation (17) is uniform in i : $\log |\mathcal{E}_i| \leq \log N_{\max}$ for every epoch. This lets us pull the $\log |\mathcal{E}_i|$ factor in Step 5 outside the sum over epochs.

Step 8 (Summing across epochs). Sum Equation (14) over $i = 1, \dots, K'$ and add Equation (16). The algorithm’s total expected utility is then at least the benchmark’s total utility, minus the discovery term $2K_{\max}$, minus $2\epsilon T$ from the telescoping Lemma-4.3 approximation errors (using $\sum_i L_i \leq T$), minus the sum of the per-epoch PW regret terms. Rearranging,

$$\mathbb{E}[\text{Regret}(T)] \leq 2K_{\max} + 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=1}^{K'} \sqrt{L_i}. \quad (18)$$

Step 9 (Cauchy–Schwarz). The last term involves the sum of K' square roots. By the Cauchy–Schwarz inequality applied to $(1, \dots, 1) \in \mathbb{R}^{K'}$ and $(\sqrt{L_1}, \dots, \sqrt{L_{K'}})$,

$$\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \leq \sqrt{K_{\max} T}. \quad (19)$$

The first inequality is tight when all epochs have equal length, a scenario that corresponds to an adversary who reveals types evenly across the horizon. This is where the leading multiplicative $\sqrt{K_{\max}}$ factor in the bound is generated. An instance-adaptive version uses K' in place of $K_{\max}$, which is what Corollary 8 records.

Step 10 (Final substitution). Plugging Equations (17) and (19) into Equation (18) and choosing $\epsilon = 1/\sqrt{T}$ gives

$$\begin{aligned} \mathbb{E}[\text{Regret}(T)] &\leq 2K_{\max} + 2\sqrt{T} + c_1 \sqrt{K_{\max} T \log N_{\max}} \\ &\leq 2K_{\max} + 2\sqrt{T} \\ &\quad + c_3 \sqrt{TK_{\max}(n^2 + K_{\max} \log n)} \end{aligned}$$

for an absolute constant c_3 . Whenever $T \geq K_{\max}$ (the only regime in which the statement of Theorem 6 is non-trivial) the leading $\sqrt{TK_{\max}(n^2 + K_{\max} \log n)}$ term dominates the additive $2K_{\max} + 2\sqrt{T}$, so

$$\mathbb{E}[\text{Regret}(T)] = O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right),$$

which is exactly the bound of Theorem 6. This completes the proof. $\square$

6.1 Consequences of the proof

Two corollaries follow directly from the proof of Theorem 6 above. The first characterizes the regime in which the bound is non-trivial, the second gives an instance-adaptive refinement.

Corollary 7 (Regime of applicability). *The bound of Theorem 6 is sublinear in T if and only if $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$; equivalently, $K_{\max} = o(\min\{T/n^2, \sqrt{T/\log n}\})$. In particular, for $K_{\max} = O(1)$ the rate is $\tilde{O}(n\sqrt{T})$.*

Proof. Set the right-hand side of Equation (7) equal to T and solve for $K_{\max}$. The condition $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$ is exactly when the bound of Theorem 6 is $o(T)$; the equivalent form follows by separately bounding $K_{\max} \cdot n^2 \leq T/\omega(1)$ and $K_{\max}^2 \log n \leq T/\omega(1)$. $\square$

Corollary 8 (Instance-adaptive bound). *Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ be the number of distinct attacker types actually observed in the realized sequence. Then Algorithm 1 achieves*

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K' \cdot (n^2 + K' \log n)}\right).$$

Proof. The Cauchy–Schwarz step Equation (19) of the proof of Theorem 6 gives $\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot T}$; plugging this sharper bound into Equation (18) and redoing Step 10 with K' in place of $K_{\max}$ gives the stated bound. $\square$

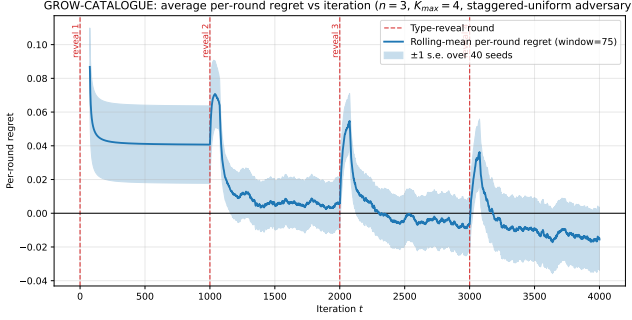


Figure 1: Rolling-window per-round regret of GROW-CATALOGUE under the staggered-uniform adversary ($n = 3$, $K_{\max} = 4$, $T = 4000$, average over 40 seeds, window = 75). Red dashed lines mark the discovery rounds 0, 1000, 2000, 3000. Each reveal causes a visible transient spike, followed by PW re-convergence inside the new epoch.

7 Experiments

7.1 Setup

We implement Algorithm 1 in Python, along with two reference algorithms: the known- K oracle of Balcan et al. [2, Sec. 5], which receives the full set K at $t = 1$; and a *uniform* baseline that plays $\mathbf{p}_{\text{def}} = (R/n, \dots, R/n)$ every round. Unless stated otherwise, we use $n = 3$ targets, $R = 1$ resource, and attacker utilities drawn i.i.d. from $\text{Unif}[-1, 1]$ with $u_{\alpha}^u(i) > 0 > u_{\alpha}^c(i)$ to ensure covering is costly for the attacker. The extreme-point set $\mathcal{E}(C; \epsilon)$ is computed by direct enumeration of intersections of $(n - 1)$ -subsets of the defining hyperplanes, which is tractable at $n = 3$.

We use a *staggered-uniform* adversary: the horizon $[0, T]$ is partitioned into $K_{\max}$ phases of length $T/K_{\max}$ each, and in phase i the attacker is drawn uniformly at random from the first i types $\{\alpha_1, \dots, \alpha_i\}$. Thus α_1 is present in every phase, α_2 from phase 2 onward, and so on. This design produces a clean sequence of discovery rounds at $t = 0, T/K_{\max}, \dots, (K_{\max} - 1)T/K_{\max}$, while still mixing old and new types after each reveal so that the benchmark $\mathbf{p}^*$ is a genuine compromise rather than a specialist strategy.

7.2 Results

Figure 1 shows that the algorithm’s per-round regret tracks the epoch structure of the proof: each discovery round produces a bounded transient, and the PW sub-routine re-converges within the new epoch. The spikes shrink with i because, by epoch i , the algorithm already has a good PW distribution on most of the new (refined) extreme points.

Figure 2 plots cumulative $\text{Regret}(T)$ versus T with the $C\sqrt{T}$ envelope from Theorem 6 overlaid. The empirical regret stays visibly below the envelope for all T tested, and the ratio $\text{Regret}(T)/\sqrt{T}$ remains bounded (in fact, it decreases, indicating that the stochastic staggered-uniform adversary is far from the worst case). This is entirely consistent with Theorem 6, which is our upper bound.

Table 1 compares the four algorithms at $T = 4000$. The regret of GROW-CATALOGUE is within roughly 15%–40% of

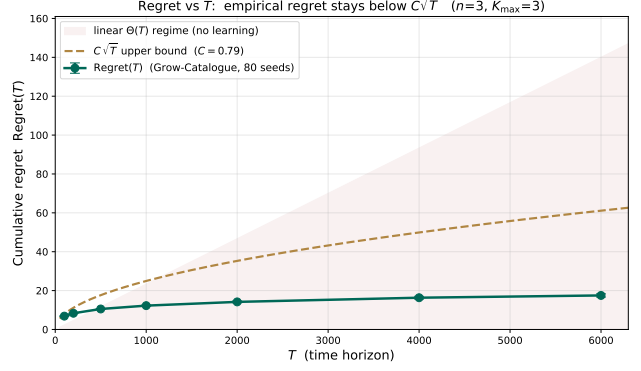


Figure 2: Cumulative regret versus horizon T for GROW-CATALOGUE ($n = 3$, $K_{\max} = 3$, uniform random adversary, 80 seeds). The green curve stays strictly below the $C\sqrt{T}$ upper envelope implied by Theorem 6, which is in turn well below the linear-regret regime that would result without learning.

the known- K oracle and an order of magnitude better than the uniform baseline. The two GROW-CATALOGUE variants produce regret that differs only by Monte-Carlo noise, as predicted by the observational equivalence of the two recomputation schemes.

Algorithm	$\mathbb{E}[\text{Regret}(T)]$	Ratio
Known- K oracle of Balcan et al. [2]	14.51 ± 5.00	1.00
GROW-CAT. (total)	16.57 ± 5.40	1.14
Uniform baseline	292.77 ± 350.28	20.18

Table 1: Cumulative regret at $T = 4000$, $n = 3$, $K_{\max} = 3$, uniform-random adversary, mean $\pm$ std over 15 seeds. The known- K oracle receives K in advance; GROW-CATALOGUE attains comparable regret *without* this information.

8 Conclusions

We studied the repeated Stackelberg security game in a new, open regime in which the set of attacker types is unknown to the defender a priori but known to have bounded cardinality. We presented GROW-CATALOGUE, a simple online algorithm that incrementally discovers attacker types and re-partitions the coverage polytope accordingly, and we gave a full proof that its expected regret under full-information feedback is at most $O(\sqrt{T} \cdot K_{\max} \cdot (n^2 + K_{\max} \log n))$. This is the same functional form as the known- K bound of Balcan et al. [2, Theorem 5.1], with the known cardinality k replaced by our known upper bound $K_{\max}$. The bound is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$, and the matching $\Omega(T)$ lower bound of Balcan et al. [2, Theorem 7.1] rules out further improvements on the unbounded side. Our experiments confirm the analytical scaling: the empirical regret stays below the $C\sqrt{T}$ envelope and tracks the epoch structure of the proof.

9 Partial-Information Feedback with an Unknown Type Set

Theorem 6 shows that not knowing K is affordable: under full-information feedback it costs a factor $\sqrt{K_{\max}}$ over the known- K rate of Balcan et al. [2, Theorem 5.1]. Not observing a_t is affordable too: with K known and only the attacked target observed, Balcan et al. [2, Theorem 6.1] achieve $O(T^{2/3}nk \log^{1/3}(nk))$. One expects the two relaxations to compose. They do not.

We show that under partial information with an unknown type set no algorithm achieves regret $o(T)$ — *even when* $K_{\max} = 1$, so that one fixed type attacks in every round. The obstruction is not how many types the adversary has, but that the defender can never learn *which* type it faces: it sees only a target index, and infinitely many types produce that same index. The diagnosis is that the defender is missing a *name*, and a name is a bounded object — which is what Section 10 then supplies. Proposition 14 closes the section by showing that this naming account is special to $K_{\max} = 1$: two types already force $\Omega(\sqrt{T})$ on a two-element universe, under full information and with K known.

9.1 The problem

Problem 9 (Unknown- K repeated SSG, partial information). A repeated SSG as in Section 3 is played, with three modifications.

1. There is a set $\mathcal{U}$ of attacker types, the *universe*, known to the defender and not assumed finite.
2. The set $K \subseteq \mathcal{U}$ is not revealed to the defender; it is given only a constant $K_{\max} \geq 1$ with the promise $|K| \leq K_{\max}$.
3. Feedback is *partial information* [2, Section 3]: at the end of round t the defender observes only the attacked target $b_{a_t}(\mathbf{p}_t)$. It never observes a_t , and utility vectors are never revealed.

The adversary is oblivious, and $\mathbf{p}^*$ and $\text{Regret}(T)$ are as in Equations (3) and (4).

The defender’s own payoff is not extra feedback: it knows u_d^c and u_d^u , so $U_d(i_t, \mathbf{p}_t)$ follows from the observed target and its own play. And Problem 9 is at least as hard as each setting it weakens: revealing a_t recovers Problem 4, and revealing K at $t = 0$ recovers Balcan et al. [2, Section 6].

9.2 The gadget G_3

Every lower bound below is proved in one fixed game G_3 : the game of Balcan et al. [2, Theorem 7.1], with $n = 3$ targets, one resource, and schedules $\mathcal{D} = \{\emptyset, \{1\}, \{2\}, \{3\}\}$, so that $\mathcal{P} = \{\mathbf{p} \in [0, 1]^3 : p_1 + p_2 + p_3 \leq 1\}$. Letting the resource idle only enlarges the defender’s action set and so only strengthens the lower bounds; every coverage vector we exhibit has $\sum_i p_i = 1$, so each argument also runs verbatim on the simplex. The defender’s payoffs are

$$u_d^c(i) = u_d^u(i) = -1 \quad (i \in \{1, 3\}), \quad u_d^c(2) = u_d^u(2) = 0, \quad (20)$$

so that by Equation (1)

$$U_d(1, \mathbf{p}) = U_d(3, \mathbf{p}) = -1, \quad U_d(2, \mathbf{p}) = 0 \quad (21)$$

for every $\mathbf{p} \in \mathcal{P}$. The defender’s payoff therefore depends on $\mathbf{p}_t$ only through which target the attacker picks: a round is worth 0 if target 2 is attacked and -1 otherwise. We say the defender *hits* at round t if $b_{a_t}(\mathbf{p}_t) = 2$ and *misses* otherwise. Types break ties towards the lowest-indexed target, as in Balcan et al. [2, Section 7]; at every benchmark we exhibit, target 2 is the *strict* maximizer, so the benchmark’s value does not depend on the rule.

Threshold types. For $0 \leq r_1 \leq r_2 \leq 1$ let α_{r_1, r_2} be the attacker with

$$u_\alpha^c = (r_1 - 1, 0, -r_2), \quad u_\alpha^u = (r_1, 0, 1 - r_2), \quad (22)$$

all six entries of which lie in $[-1, 1]$. Substituting into $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1 - p_i)$ gives, for every $\mathbf{p} \in \mathcal{P}$,

$$U_\alpha(1, \mathbf{p}) = r_1 - p_1, \quad U_\alpha(2, \mathbf{p}) = 0, \quad U_\alpha(3, \mathbf{p}) = (1 - r_2) - p_3. \quad (23)$$

So on the segment $\mathbf{p} = (p, 0, 1 - p)$ the three utilities are $r_1 - p$, 0 and $p - r_2$, and comparing them gives

$$b_{\alpha_{r_1, r_2}}((p, 0, 1 - p)) = \begin{cases} 1 & p \in [0, r_1], \\ 2 & p \in (r_1, r_2], \\ 3 & p \in (r_2, 1]. \end{cases} \quad (24)$$

This is Balcan et al. [2, Lemma 7.2]; we record the utility vectors explicitly because the arguments below need the best response at every $\mathbf{p} \in \mathcal{P}$, not only on that segment. A type is thus named by an interval $(r_1, r_2]$, and the defender’s task is to locate it.

The next lemma is the structural fact driving every lower bound: a hit is very informative but very rare, because one coverage vector can hit at most one type of any disjoint family.

Lemma 10 (the hit condition). *Let $0 \leq r_1 < r_2 \leq 1$ and $\alpha = \alpha_{r_1, r_2}$. For every $\mathbf{p} \in \mathcal{P}$,*

$$b_\alpha(\mathbf{p}) = 2 \iff p_1 > r_1 \text{ and } p_3 \geq 1 - r_2, \quad (25)$$

and then $p_1 \in (r_1, r_2]$. Hence if the intervals of a family of types are pairwise disjoint, every $\mathbf{p} \in \mathcal{P}$ hits at most one of them.

Proof. By Equation (23) and lowest-index tie-breaking, target 2 is chosen exactly when $U_\alpha(2, \mathbf{p}) = 0$ strictly beats $U_\alpha(1, \mathbf{p}) = r_1 - p_1$ and weakly beats $U_\alpha(3, \mathbf{p}) = (1 - r_2) - p_3$, which is Equation (25). Then $p_1 \leq 1 - p_2 - p_3 \leq 1 - p_3 \leq r_2$, so $p_1 \in (r_1, r_2]$; and a single p_1 cannot lie in two disjoint intervals. $\square$

9.3 Identification is a comparison search

Fix $M \geq 2$ and let

$$I_j = \left(\frac{j-1}{M}, \frac{j}{M} \right], \quad \alpha^{(j)} = \alpha_{\frac{j-1}{M}, \frac{j}{M}}, \quad j \in [M], \quad (26)$$

so $I_1, \dots, I_M$ partition $(0, 1]$ and $\mathcal{U}_M := \{\alpha^{(1)}, \dots, \alpha^{(M)}\}$ consists of M distinct types, to which Lemma 10 applies.

Facing an instance $K = \{\alpha^{(j)}\}$, the defender is searching for the index j : each play is a query, each observation an answer in $\{1, 2, 3\}$, and by Lemma 10 the answer “2” is consistent with at most one candidate. The following lemma bounds how

fast *any* such search can succeed. It sharpens the classical $\Omega(\log_2 M)$ comparison-search bound in the two ways we need: it charges every unsuccessful query, not just the number of queries until termination, and it bounds the whole tail of the success time rather than only its worst case.

Lemma 11 (comparison trees identify slowly). *Let $M \geq 1$ and let j be uniform on $[M]$. Consider any interactive protocol in which, for $t = 1, 2, \dots$, a deterministic strategy picks a query q_t as a function of everything seen so far and receives an answer $o_t = \mathcal{O}(q_t, j) \in \{1, 2, 3\}$. Assume only that*

$$|\{j' \in [M] : \mathcal{O}(q, j') = 2\}| \leq 1 \quad \text{for every query } q. \quad (27)$$

Let $\tau = \min\{t : o_t = 2\}$, with $\tau = \infty$ if there is no such t . Then for every $u \geq 1$ and every $T \geq 1$,

$$\Pr[\tau \leq u] \leq \frac{2^u - 1}{M}, \quad (28)$$

$$\mathbb{E}[\min\{\tau - 1, T\}] \geq T - \frac{2^{T+1} - T - 2}{M}. \quad (29)$$

Proof. The idea is a counting one: a searcher that has not yet hit has learned only which of 2^{t-1} answer histories it is in, and each such history can be the prefix of a hit for at most one index.

Formally, let H_t be the set of histories $(o_1, \dots, o_{t-1})$ of the first $t-1$ rounds containing no answer 2; each $o_s \in \{1, 3\}$, so $|H_t| \leq 2^{t-1}$. If $\tau(j) = t$ then j produces some $h \in H_t$; the strategy being deterministic, the round- t query is the fixed $q_t(h)$, and $\mathcal{O}(q_t(h), j) = 2$. By Equation (27) at most one index answers 2 to $q_t(h)$, so each h accounts for at most one such j , giving $|\{j : \tau(j) = t\}| \leq 2^{t-1}$. Summing over $t \leq u$ and dividing by M gives Equation (28).

For Equation (29), put $X = \min\{\tau - 1, T\}$. Since $X > s$ iff $\tau > s + 1$,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{s=0}^{T-1} \Pr[X > s] = \sum_{u=1}^T (1 - \Pr[\tau \leq u]) \\ &\geq T - \sum_{u=1}^T \frac{2^u - 1}{M}, \end{aligned}$$

which is Equation (29) because $\sum_{u \leq T} 2^u = 2^{T+1} - 2$. $\square$

9.4 The impossibility theorem

Theorem 12 (unknown K with partial information is not learnable). *Let G_3 be as in Equation (20) and let $\mathcal{U} = \{\alpha_{r_1, r_2} : 0 \leq r_1 \leq r_2 \leq 1\}$ be the infinite universe of threshold types Equation (22). For every $T \geq 1$ and every $K_{\max} \geq 1$, every online algorithm for Problem 9 — randomized or deterministic, even one knowing T , $\mathcal{U}$, $K_{\max}$ and the construction — suffers, on some instance with $|K| = 1$,*

$$\mathbb{E}[\text{Regret}(T)] \geq T - 2 + \frac{T+2}{2^T} > T - 2. \quad (30)$$

The game does not depend on T ; only the adversary's choice of K inside the fixed universe does.

Proof. Set $M = 2^T$ and take $\mathcal{U}_M \subset \mathcal{U}$ as in Equation (26). By Yao's minimax principle it suffices to exhibit a distribution over instances against which every *deterministic* algorithm suffers this much: draw j uniformly from $[M]$ and set $K = \{\alpha^{(j)}\}$, with $a_t = \alpha^{(j)}$ for all t . This is legal for every $K_{\max} \geq 1$ and oblivious, since $\mathbf{a}$ is fixed before play.

The benchmark is 0. Let $\mathbf{p}^* = (\frac{2j-1}{2M}, 0, 1 - \frac{2j-1}{2M})$, whose first coordinate is the midpoint of I_j . Then $p_1^* > r_1$ and $p_3^* > 1 - r_2$ strictly, so $b_{\alpha^{(j)}}(\mathbf{p}^*) = 2$ by Equation (25) under any tie-breaking rule, and $\mathbf{p}^*$ earns 0 every round. As $U_d \leq 0$ everywhere by Equation (21), no coverage vector does better, so $\mathbf{p}^*$ is a maximizer in Equation (3) and

$$\text{Regret}(T) = \mathbb{E}[\#\{t \leq T : \text{miss at } t\}]. \quad (31)$$

Regret counts missed rounds exactly.

The interaction is a comparison search. A deterministic algorithm is a family of maps $\mathbf{p}_t = \mathbf{p}_t(o_1, \dots, o_{t-1})$, the observations being its only feedback. Take $q_t = \mathbf{p}_t$ and $\mathcal{O}(\mathbf{p}, j') = b_{\alpha^{(j')}}(\mathbf{p})$. The I_j are disjoint, so Lemma 10 gives exactly Equation (27).

Counting. Every round before the first hit τ is a miss, so by Equation (31) and Equation (29) with $M = 2^T$,

$$\mathbb{E}_j[\text{Regret}(T)] \geq \mathbb{E}_j[\min\{\tau - 1, T\}] \geq T - \frac{2^{T+1} - T - 2}{2^T},$$

which is Equation (30). Averaging over a randomized algorithm's internal randomness preserves the bound, since it holds for every deterministic realization. $\square$

Remark 13 (the bound is exactly tight). The constant cannot be improved: the defender that keeps the set of indices consistent with its observations — an integer interval, by Equation (24) — and plays $\mathbf{p}_t = (k/M, 0, 1 - k/M)$ for its midpoint k attains expected regret exactly $T - 2 + (T+2)2^{-T}$, as Section 11.1 verifies. Since $\text{Regret}(T) \leq T$ always in G_3 , the defender loses all but two of the T rounds.

Letchford et al. [13] show that a defender facing a *single* unknown attacker type can learn a near-optimal commitment from a small number of best-response queries, which at first sight is in tension with Theorem 12. There is none: any such bound must depend on the geometry of the best-response regions, and the family $\mathcal{U}_M$ of Equation (26) drives the region that has to be located down to width $1/M = 2^{-T}$.

9.5 A $\sqrt{T}$ floor as soon as $K_{\max} \geq 2$

For $K_{\max} = 1$ the problem is entirely a naming problem. On a fixed finite universe $\mathcal{U}_M$ the version-space search of Remark 13 finishes after at most $\lceil \log_2 M \rceil$ misses and hits in every round thereafter, so the regret stops growing: it does not grow with T at all, and the linear bound of Theorem 12 is driven purely by $|\mathcal{U}| = \infty$. That is special to one type. With two types the regret is $\Omega(\sqrt{T})$ on the *smallest possible* universe, under *full* information, and even if the defender is told K in advance — because it must now track which of two known opponents is currently more frequent, and no naming settles that.

Proposition 14 (two types already cost $\sqrt{T}$). *Consider G_3 with the two types $\alpha^{(1)} = \alpha_{0, 1/2}$ and $\alpha^{(2)} = \alpha_{1/2, 1}$, so $|\mathcal{U}| = |K| = K_{\max} = 2$, and let $a_1, \dots, a_T$ be drawn i.i.d.*

uniformly from K . Then every online algorithm — randomized or deterministic, under either feedback model, and even one told K with its utility vectors at $t = 0$ — satisfies

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{1}{2}\sqrt{[T/2]} \quad (32)$$

for every $T \geq 2$; consequently some fixed sequence forces at least this much.

Proof. The two intervals $(0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ are disjoint, so by Lemma 10 every $\mathbf{p}$ hits at most one type. Write $N_i = |\{t : a_t = \alpha^{(i)}\}|$, so $N_1 + N_2 = T$.

Benchmark. A fixed $\mathbf{p}$ earns 0 when it hits and -1 otherwise, so it earns at least $-\min\{N_1, N_2\}$, and this is attained: $\mathbf{p} = (\frac{1}{4}, 0, \frac{3}{4})$ hits $\alpha^{(1)}$ in all N_1 of its rounds by Equation (25), and $(\frac{3}{4}, 0, \frac{1}{4})$ hits $\alpha^{(2)}$ in all N_2 of its rounds. Hence the benchmark earns $-\min\{N_1, N_2\}$.

Algorithm. Fix t and condition on the history, which depends only on $a_1, \dots, a_{t-1}$ and the algorithm's randomness, hence is independent of a_t . The played $\mathbf{p}_t$ hits at most one type, and a_t is that type with probability $\frac{1}{2}$, so $\Pr[\text{hit at } t] \leq \frac{1}{2}$ and the algorithm earns at most $-T/2$ in expectation.

Combining. As $\min\{N_1, N_2\} = \frac{1}{2}(T - |N_1 - N_2|)$,

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{T}{2} - \mathbb{E}[\min\{N_1, N_2\}] = \frac{1}{2}\mathbb{E}[|N_1 - N_2|],$$

and $N_1 - N_2$ is a simple random walk S_T , for which $\mathbb{E}|S_T| \geq \sqrt{[T/2]}$ (Appendix A). Nothing here refers to the feedback model or to knowledge of K : only that $\mathbf{p}_t$ cannot hit both types and that a_t is independent of the past. $\square$

The bound is tight on this instance in a strong sense: $\Pr[\text{hit}] \leq \frac{1}{2}$ is an *equality* for any algorithm that plays inside one of the two intervals, so every such algorithm — follow-the-leader included, and one simply told K included — has expected regret exactly $\frac{1}{2}\mathbb{E}|N_1 - N_2| = \sqrt{T/(2\pi)}(1 + o(1))$. The remaining freedom buys nothing; Figure 3(b) shows the measured curve sitting on that value.

So the transition at $K_{\max} = 2$ is not about how hard a type is to identify but about whether identification is the whole problem. On the upper side, Balcan et al. [2, Theorem 6.1] with $k = M$ — legitimate, since a finite universe is a known type set — gives $O(\min\{T, T^{2/3}nM \log^{1/3}(nM)\})$ under partial information, and Theorem 6 gives $O(\sqrt{TK_{\max}(n^2 + K_{\max} \log n)})$ under full information. Closing the gap to $\Omega(\sqrt{T})$ is the finite- M question of Section 12.

10 Block Revelation: Recovering Sublinear Regret

Theorem 12 says the defender must be told *something* about who its opponents are, and it says exactly what is missing: not which types attack when, but the *identity* of the types — a name. A name is a bounded object, so this is a supplement one can actually buy. This section supplies the weakest such supplement we could find that is realistic in the applications of Section 2 — a periodic threat-intelligence report — and shows it restores sublinear regret at the partial-information rate of Balcan et al. [2, Theorem 6.1].

The supplement is deliberately meagre: every B rounds an oracle hands over the *set* of types that have attacked so far,

with their utility vectors. It never says when any of them attacked, how often, or how many are still to come, so the defender still cannot attribute a round to a type — precisely the difficulty Balcan et al. [2, Section 6] solve by estimating frequencies from best responses. That difficulty is untouched, which is why the rate below is $T^{2/3}$ and not $\sqrt{T}$.

10.1 The model

Problem 15 (Unknown- K repeated SSG with block revelation). Fix a *revelation length* $B \in \mathbb{N}$, known to the defender, and split the horizon into $N_B = \lceil T/B \rceil$ *revelation blocks*

$$\mathcal{B}_\nu = \{(\nu - 1)B + 1, \dots, \min\{\nu B, T\}\}, \quad \nu \in [N_B]. \quad (33)$$

The game is Problem 9 with one addition: at the end of $\mathcal{B}_\nu$ the defender receives

$$D_\nu = \{a_t : t \leq \min\{\nu B, T\}\} \subseteq K \quad (34)$$

together with the utility vectors of each $\alpha \in D_\nu$, where $D_0 = \emptyset$. Nothing else is revealed — not $|K|$, not when any type attacked, not how often. Within a round the feedback is still partial.

Three remarks delimit the model. It is strictly weaker than Problem 4: even at $B = 1$ the defender learns a round's type only when that type is new. Taking $B \geq T$ makes the revelation useless and recovers Problem 9, so by Theorem 12 some bound on B is necessary — Theorem 22 quantifies which. And the oracle is cumulative rather than incremental, but a defender that remembers $D_{\nu-1}$ can convert between the two, so we use whichever is convenient.

10.2 The algorithm

Algorithm 2 is Algorithm 1 with two changes forced by the model: the catalogue is refreshed at checkpoints rather than at discovery rounds, and the inner learner is the partial-information algorithm of Balcan et al. [2, Algorithm 1] rather than Polynomial Weights, since an expert's loss can no longer be computed from the feedback. Write $\mathcal{A}(C)$ for a fresh instance run on $\mathcal{E}(C; \epsilon)$ with attacker set C ; it spends $|C|$ rounds of each internal estimation block playing barycentric-spanner strategies and the rest playing what its full-information subroutine proposes. We fix $\epsilon = 1/T$ throughout rather than tuning it per segment: this is admissible because the discretization error of Lemma 1 is $2\epsilon L$ and only decreases with ϵ , while the size bound of Lemma 2 does not involve ϵ at all.

One property that Theorem 6 got for free must be arranged here. There the inner learner was Polynomial Weights, which is anytime, so an epoch's length being revealed only at its end cost nothing; the algorithm of Balcan et al. [2, Theorem 6.1] instead tunes itself to a horizon given in advance. Lemma 26 in Appendix B supplies the anytime form we need: there is an $\mathcal{A}(C)$ not receiving its horizon, and an absolute constant c_4 , whose regret on $[1, L]$ is at most $c_4 L^{2/3} n |C| \log^{1/3}(n |C|)$ for every $L \leq T$ simultaneously, against any oblivious sequence drawn from C . Both features matter below: the guarantee must hold at every prefix, because the instance keeps running past the epoch we bound.

Algorithm 2 BLOCK-GROW-CATALOGUE (partial information, revelation length B).

Require: horizon T , revelation length B , game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

```

1:  $C \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/T$ ;  $\mathcal{A} \leftarrow \text{NULL}$ 
2: for  $t = 1, \dots, T$  do
3:   if  $C = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     play the  $\mathbf{p}_t$  prescribed by  $\mathcal{A}$ 
7:   end if
8:   observe the attacked target  $i_t = b_{a_t}(\mathbf{p}_t)$ ; pass  $i_t$  to  $\mathcal{A}$ 
9:   if  $t = \nu B$  for some  $\nu$ , or  $t = T$  then ▷ checkpoint
10:    query the oracle for  $D_\nu$ 
11:    if  $D_\nu \neq C$  then ▷ discovery block
12:       $C \leftarrow D_\nu$ 
13:       $\mathcal{A} \leftarrow \mathcal{A}(C)$  ▷ restart; the state built during  $\mathcal{B}_\nu$  is discarded
14:    end if
15:  end if
16: end for

```

10.3 Block structure

The analysis rests on a decomposition of the horizon playing the role of Step 1 of Section 6, but at the granularity of blocks. The picture is simple: blocks in which the revealed set grows are written off, and between them lie *epochs* on which the catalogue is fixed and — crucially — correct.

Definition 16 (discovery blocks and epochs). Call $\mathcal{B}_\nu$ a *discovery block* if $D_\nu \supsetneq D_{\nu-1}$ and a *clean block* otherwise, and let $\beta_1 < \dots < \beta_m$ enumerate the discovery blocks. For $j \in [m]$ the j -th *epoch* is the set of rounds in the clean blocks separating β_j from the next discovery block,

$$\text{Ep}_j = \bigcup_{\nu=\beta_j+1}^{\beta_{j+1}-1} \mathcal{B}_\nu, \quad L_j = |\text{Ep}_j|, \quad (35)$$

with $\beta_{m+1} := N_B + 1$.

Lemma 17 (the decomposition). *Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ be the number of types that actually appear, and let $C_j := D_{\beta_j}$ be the catalogue Algorithm 2 holds at the end of block β_j . Then*

1. $\beta_1 = 1$ and $m \leq K'$, and $[1, T] = (\bigsqcup_j \mathcal{B}_{\beta_j}) \sqcup (\bigsqcup_j \text{Ep}_j)$ with the discovery blocks contributing at most $mB \leq K'B$ rounds and $\sum_j L_j \leq T$;
2. (containment) if $\mathcal{B}_\nu$ is clean then $a_t \in D_{\nu-1}$ for every $t \in \mathcal{B}_\nu$; consequently $C_t = C_j$ and $a_t \in C_j$ for every $t \in \text{Ep}_j$, and throughout Ep_j the algorithm runs one uninterrupted instance $\mathcal{A}(C_j)$ started at the first round of Ep_j , no estimate of which at a round of Ep_j rests on a type outside C_j ;
3. (determinism) $\beta_1, \dots, \beta_m$ and $\text{Ep}_1, \dots, \text{Ep}_m$ are functions of $\mathbf{a}$ alone, independent of the defender's plays and randomization.

Proof. (i) $D_0 = \emptyset \neq D_1$, so $\mathcal{B}_1$ is a discovery block. Each discovery block adds a type, so $m \leq |D_{\beta_m}| \leq K'$. Every block is either a discovery block or lies in exactly one Ep_j , and each holds at most B rounds; the Ep_j are disjoint subsets of $[1, T]$.

(ii) Contrapositive of the first claim: if $a_t \notin D_{\nu-1}$ for some $t \in \mathcal{B}_\nu$ then $a_t \in D_\nu \setminus D_{\nu-1}$ by Equation (34), so $\mathcal{B}_\nu$ is a

discovery block. For the rest, Algorithm 2 changes C only at discovery blocks, so C is constant on Ep_j and equals D_{β_j} ; and for $t \in \mathcal{B}_\nu \subseteq \text{Ep}_j$ we get $a_t \in D_{\nu-1} = D_{\beta_j} = C_j$ since no block strictly between β_j and ν is a discovery block. No restart is triggered inside Ep_j , and the instance was restarted at the checkpoint closing β_j .

The restriction to rounds of Ep_j is not cosmetic: the instance is *not* discarded when Ep_j ends. It runs on through the next discovery block, which is exactly where a contaminated round can occur; Algorithm 2 discards it at the checkpoint closing that block, so no contaminated observation reaches a later epoch, and since Theorem 18 uses only the prefix Ep_j , none enters the bound at all.

(iii) By Equation (34) each D_ν is a function of $a_1, \dots, a_{\min\{\nu B, T\}}$, hence so is the predicate “ $\mathcal{B}_\nu$ is a discovery block”, hence so are the β_j and the epochs. $\square$

Part (ii) is what makes the analysis work, and it deserves a sentence. The defender cannot tell, *while a block is being played*, whether the attacking type is one it knows. It does not have to: a round whose type is outside the catalogue can only sit inside a block that the very next checkpoint flags as a discovery block. Contamination is confined to blocks that get detected — one checkpoint late, but before any of it can influence a later epoch.

10.4 Main result

Theorem 18 (block revelation restores no-regret). *Let the defender play Algorithm 2 in Problem 15. For any oblivious $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$, and $K' \leq K_{\max}$ the number of distinct types appearing,*

$$\mathbb{E}[\text{Regret}(T)] \leq c n K'^{4/3} T^{2/3} \log^{1/3}(nK') + 2K'B \quad (36)$$

for an absolute constant c ; in particular $\mathbb{E}[\text{Regret}(T)] = O(nK_{\max}^{4/3} T^{2/3} \log^{1/3}(nK_{\max}) + K_{\max}B)$.

Proof. Fix the oblivious sequence $\mathbf{a}$ and adopt the notation of Definition 16 and Lemma 17. Throughout, for a set of rounds $S \subseteq [1, T]$ write

$$\text{Regret}(S) = \sum_{t \in S} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in S} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \quad (37)$$

for the regret accumulated on S against the global benchmark, so that $\text{Regret}([1, T]) = \text{Regret}(T)$ and $\text{Regret}(\cdot)$ is additive over disjoint sets. The proof follows the ten steps of Section 6; the ones that differ are Steps 1, 3 and 6.

Step 1 (Block decomposition). By Lemma 17(i) the horizon is the disjoint union of the $m \leq K'$ discovery blocks and the m epochs, so by additivity of Equation (37)

$$\text{Regret}(T) = \sum_{j=1}^m \text{Regret}(\text{Ep}_j) + \sum_{j=1}^m \text{Regret}(\mathcal{B}_{\beta_j}). \quad (38)$$

Unlike Step 1 of Section 6, where consecutive epochs were separated by *single* discovery rounds, here they are separated by whole blocks. That is the only structural change, and it is what produces the additive B .

Step 2 (Catalogue is constant and correct inside an epoch). This is Lemma 17(ii): $C_t = C_j$ and $a_t \in C_j$ for every $t \in \text{Ep}_j$, and throughout Ep_j the defender runs one uninterrupted instance $\mathcal{A}(C_j)$ started at the first round of Ep_j . It is the exact analogue of Equation (9).

Step 3 (Within-epoch regret). Fix j . By Step 2, $\mathcal{A}(C_j)$ — the anytime wrapper of Lemma 26, not the horizon-dependent algorithm inside it — runs on a sequence of length L_j drawn from the known set C_j , under partial information, with no contaminated estimate. Two hypotheses need checking, and they are the two Lemma 26 was stated to cover. The sequence $(a_t)_{t \in \text{Ep}_j}$ must be *oblivious*, because Balcan et al. [2, Lemma 6.3] needs the attacker met at a random exploration round to be a random attacker of the block; it is, because by Lemma 17(iii) the epoch is a function of $\mathbf{a}$ alone. And L_j is unknown when the instance starts, which is what the anytime form handles. Hence

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[\sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_4 L_j^{2/3} n |C_j| \log^{1/3}(n |C_j|). \end{aligned} \quad (39)$$

In contrast to Equation (10), the left-hand side already compares against the unrestricted maximum over $\mathcal{P}$: Balcan et al. [2, Theorem 6.1] states its bound against the best fixed strategy in hindsight, so the passage through $\mathcal{E}(C_j; \epsilon)$ and the $2\epsilon L_j$ term of Lemma 1 are internal to their result and are not repeated here.

Step 4 (Pass from the epoch optimum to $\mathbf{p}^*$). Exactly as in Sub-step 4b of Section 6: $\mathbf{p}^* \in \mathcal{P}$, so it is a feasible competitor in the maximum on the left of Equation (39),

$$\max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \geq \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*), \quad (40)$$

and chaining Equations (39) and (40) with Equation (37) gives the per-epoch bound

$$\text{Regret}(\text{Ep}_j) \leq c_4 L_j^{2/3} n |C_j| \log^{1/3}(n |C_j|). \quad (41)$$

Step 5 (Discovery blocks). Since $U_d(i, \mathbf{p}) \in [-1, 1]$ for every target and every $\mathbf{p} \in \mathcal{P}$, exactly as in Equation (15) each round contributes at most 2 to Equation (37). A block holds at most B rounds, so $\text{Regret}(\mathcal{B}_{\beta_j}) \leq 2B$ and, by Lemma 17(i),

$$\sum_{j=1}^m \text{Regret}(\mathcal{B}_{\beta_j}) \leq 2mB \leq 2K'B. \quad (42)$$

This is the entire price of batching, charged at face value: we make no attempt to learn inside a discovery block, and Theorem 22 shows that nothing is lost by not trying.

Step 6 (Uniform bound on the catalogue). $|C_j| \leq |D_{\beta_m}| \leq K'$ for every j , so $|C_j| \log^{1/3}(n |C_j|) \leq K' \log^{1/3}(n K')$ uniformly in j and the factor leaves the sum. This is the analogue of Step 7 of Section 6, and it is shorter only because Balcan et al. [2, Theorem 6.1] has already absorbed $\log |\mathcal{E}(C_j; \epsilon)|$ into its displayed form.

Step 7 (Summing across epochs). Substituting Equations (41) and (42) into Equation (38) and applying Step 6,

$$\mathbb{E}[\text{Regret}(T)] \leq c_4 n K' \log^{1/3}(n K') \sum_{j=1}^m L_j^{2/3} + 2K'B. \quad (43)$$

Step 8 (Hölder). The sum of m two-thirds powers is controlled by Hölder's inequality with exponents 3 and 3/2, applied to $(1, \dots, 1) \in \mathbb{R}^m$ and $(L_1^{2/3}, \dots, L_m^{2/3})$:

$$\sum_{j=1}^m L_j^{2/3} \leq m^{1/3} \left(\sum_{j=1}^m L_j \right)^{2/3} \leq K'^{1/3} T^{2/3}, \quad (44)$$

using $m \leq K'$ and $\sum_j L_j \leq T$ from Lemma 17(i). As with the Cauchy–Schwarz step Equation (19), equality needs all epochs equally long — an adversary spreading its reveals evenly across the horizon — and this is where the extra $K'^{1/3}$ is generated.

Step 9 (Final substitution). Substituting Equation (44) into Equation (43) gives Equation (36) with $c = c_4$, and $K' \leq K_{\max}$ gives the second form. $\square$

Remark 19 (why $B = 1$ does not recover Theorem 6). At $B = 1$ every round is its own block, so the decomposition of Definition 16 coincides *exactly* with the round decomposition of Section 6: a discovery block is a discovery round, and the epochs are the same sets. The additive terms agree too — Equation (42) reads $2K'$, which is Equation (16). Yet the leading terms do not: Theorem 6 gives $\sqrt{T} K_{\max}(n^2 + K_{\max} \log n)$ and Equation (36) gives $n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max})$.

The reason is that $B = 1$ does not restore full information. Problem 15 reveals a type's identity only when that type is *new*; on every other round the defender still sees nothing but a target and cannot say which member of its own catalogue attacked. That is precisely the attribution problem, so the inner learner must still be the frequency-estimating algorithm of Balcan et al. [2, Theorem 6.1] rather than Polynomial Weights — and it is the inner learner, not the decomposition, that fixes the exponent. The same substitution changes the epoch price in the opposite direction: Cauchy–Schwarz on K' square roots costs $\sqrt{K'}$ in Equation (19), whereas Hölder on K' two-thirds powers costs only $K'^{1/3}$ in Equation (44). Not knowing K is cheaper here; seeing less each round is dearer, and by more.

Remark 20 (the two prices, separated). Equation (36) differs from Balcan et al. [2, Theorem 6.1] at $k = K_{\max}$ in two places, of different natures. The factor $K_{\max}^{1/3}$ is the *price of not knowing* K , produced by Hölder over the epochs, and is the partial-information counterpart of the $\sqrt{K_{\max}}$ that Cauchy–Schwarz produces in Theorem 6. The additive $2K_{\max}B$ is the *price of batching*, unavoidable by Theorem 22 up to a factor $2B/(B-2)$. The exponent 2/3 is inherited untouched: block revelation supplies identities but says nothing about frequencies, which is what the barycentric-spanner estimator must reconstruct, so improving the exponent here would improve their Theorem 6.1.

It is worth seeing what exactly buys the $K_{\max}^{1/3}$, and the cleanest way is to price a *stronger* oracle. Suppose that at one round $t = P$ the defender were told all of K , including types

that have not yet attacked. The first P rounds cost at most $2P$, and thereafter the defender is in the known- K setting, giving $\mathbb{E}[\text{Regret}(T)] \leq 2P + O((T-P)^{2/3} n K_{\max} \log^{1/3}(n K_{\max}))$ with no $K_{\max}^{1/3}$ at all. So that factor is not the price of an unknown K as such; it is the price of learning K incrementally, one type at a time, which is what forces the epoch decomposition. An oracle that can name a type before it ever attacks avoids it; the oracle of Problem 15, which only reports what has already happened, cannot.

Finally, Algorithm 2 never reads $K_{\max}$ — the bound is stated in terms of the K' types that actually appear, so it is automatically instance-adaptive in the sense of Corollary 8, and the defender needs only T and B .

10.5 How coarse may the revelation be?

Corollary 21 (free batching, and the number of oracle calls). *The additive term of Equation (36) is dominated as soon as $B \leq B^*$, where*

$$\begin{aligned} B^* &= \frac{1}{2} c n K_{\max}^{1/3} T^{2/3} \log^{1/3}(n K_{\max}) \\ &= \Theta(n K_{\max}^{1/3} T^{2/3} \log^{1/3}(n K_{\max})), \end{aligned} \quad (45)$$

and then $\mathbb{E}[\text{Regret}(T)] = O(n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max}))$: batching is free. Equivalently the defender need consult the oracle only

$$N_B = \lceil T/B^* \rceil = O\left(\frac{T^{1/3}}{n K_{\max}^{1/3} \log^{1/3}(n K_{\max})}\right) \quad (46)$$

times, and this is optimal: by Theorem 22, for $N \geq K_{\max}$ no algorithm receiving N evenly spaced revelations has worst-case regret R unless

$$N \geq \frac{K_{\max} T}{R + 2K_{\max}}, \quad (47)$$

which at $R = \Theta(n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max}))$ matches Equation (46) up to a constant. Batching is free up to $\Theta(T^{1/3})$ reports and not beyond. Finally, the bound of Theorem 18 is sublinear in T if and only if $n K_{\max}^{4/3} \log^{1/3}(n K_{\max}) = o(T^{1/3})$ and $K_{\max} B = o(T)$; for $K_{\max} = O(1)$ and $B \leq B^*$ the rate is $O(n T^{2/3})$.

Proof. Solving $2K_{\max} B \leq c n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max})$ for B gives Equation (45), and Equation (46) is $\lceil T/B^* \rceil$. For Equation (47), N evenly spaced revelations is Problem 15 with $B = T/N$; the hypothesis $T \geq K_{\max} B$ of Theorem 22 reads $N \geq K_{\max}$, and that theorem then gives $R \geq K_{\max}(T/N - 2)$, which rearranges. (For $N < K_{\max}$ the same theorem with $K_{\max}$ replaced by N gives the stronger $R \geq T - 2N$, so such an N is never worth using.) The sublinearity criterion is Equation (36) at $K' = K_{\max}$, divided by T , requiring each term to vanish. $\square$

10.6 The additive term is necessary

The $\Theta(K_{\max} B)$ term is not an artefact of the analysis. The construction applies the dyadic family Equation (26) recursively inside G_3 : each new type hides *inside* the previous one, so one coverage vector is optimal against all of them — making the benchmark perfect — while each must still be located from scratch.

Theorem 22 (a matching lower bound in B). *Let G_3 and $\mathfrak{U}$ be as in Theorem 12. For every $B \geq 1$, every $K_{\max} \geq 1$ and every $T \geq K_{\max} B$, there is a distribution over instances of Problem 15 with $|K| = K_{\max}$ on which every online algorithm suffers*

$$\mathbb{E}[\text{Regret}(T)] \geq K_{\max} (B - 2). \quad (48)$$

Proof. Put $M = 2^B$, draw $k_1, \dots, k_{K_{\max}}$ independently and uniformly from $[M]$, and define nested intervals $J_0 = (0, 1] \supset J_1 \supset \dots$ by letting J_j be the k_j -th of the M equal sub-intervals of J_{j-1} :

$$J_j = \left(\ell_{j-1} + \frac{k_j - 1}{M} |J_{j-1}|, \ell_{j-1} + \frac{k_j}{M} |J_{j-1}| \right], \quad (49)$$

ℓ_{j-1} and $|J_{j-1}|$ being the left endpoint and length of J_{j-1} . Let $\alpha^{[j]}$ be the threshold type with interval J_j , put $K = \{\alpha^{[1]}, \dots, \alpha^{[K_{\max}]}\}$, and let $a_t = \alpha^{[j]}$ for $t \in \mathcal{B}_j$ and $a_t = \alpha^{[K_{\max}]}$ for $t > K_{\max} B$. The sequence is fixed once the k_j are drawn, so the adversary is oblivious, and each J_j is a *proper* sub-interval of J_{j-1} , so the $K_{\max}$ types are distinct. As before we bound deterministic algorithms and appeal to Yao.

The benchmark is 0. Let μ be the midpoint of the innermost interval $J_{K_{\max}}$ and $\mathbf{p}^* = (\mu, 0, 1 - \mu)$. Nesting gives $r_1^{[j]} \leq r_1^{[K_{\max}]} < \mu$ and $1 - r_2^{[j]} \leq 1 - r_2^{[K_{\max}]} < 1 - \mu$ for every j , so $\mathbf{p}^*$ hits every type by Equation (25), under any tie-breaking rule, and regret again counts misses.

Each block is a fresh search. When $\mathcal{B}_j$ begins, the checkpoint closing $\mathcal{B}_{j-1}$ has revealed $\alpha^{[1]}, \dots, \alpha^{[j-1]}$, hence J_{j-1} , and nothing about $\alpha^{[j]}$, which has not yet attacked. Moreover the whole transcript of $\mathcal{B}_1, \dots, \mathcal{B}_{j-1}$ is a function of $k_1, \dots, k_{j-1}$ alone: inductively, each play is determined by earlier observations, and each observation by that play and the interval then attacking. As k_j is drawn independently, it remains uniform on $[M]$ conditionally on everything the defender has seen and been told.

Condition on that transcript. Within $\mathcal{B}_j$ the defender's plays are deterministic functions of the observations made inside $\mathcal{B}_j$, the only attacker is $\alpha^{[j]}$, and its interval is one of M disjoint sub-intervals of J_{j-1} , so Lemma 10 gives Equation (27). The conditioning has absorbed everything the oracle said before $\mathcal{B}_j$, and it says nothing during $\mathcal{B}_j$, so within the block the interaction is a plain protocol of the kind Lemma 11 covers. Lemma 11 with $M = 2^B$ and horizon B gives $\mathbb{E}[\#\text{misses in } \mathcal{B}_j] \geq B - (2^{B+1} - B - 2)/2^B > B - 2$. Summing over $j \in [K_{\max}]$ and averaging over the algorithm's randomness gives Equation (48). $\square$

Remark 23 (the additive term is tight). Theorems 18 and 22 bracket the cost of batching between $K_{\max}(B-2)$ and $2K_{\max}B$, so it is $\Theta(K_{\max}B)$ for $B \geq 3$, the two differing by $2B/(B-2) - 4$ at $B = 4$ and $2 + o(1)$ thereafter; for $B \leq 2$ the lower bound is vacuous. The two endpoints are worth naming. At $K_{\max} = 1$, $B = T$ the bound reads $\mathbb{E}[\text{Regret}(T)] \geq T - 2$, so Theorem 12 is exactly the $B \rightarrow T$ limit of Theorem 22. At $B = 1$ the additive term is $2K_{\max}$ and Equation (36) becomes $O(n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max}) + K_{\max})$ — by Remark 19 the partial-information counterpart of Theorem 6, and to our knowledge the first sublinear bound for an unknown attacker set under partial-information feedback.

10.7 Scheduled versus chosen revelation times

Problem 15 fixes the checkpoints before play, which is what a subscription looks like: reports arrive on a calendar, not on demand. How much of the $\Theta(K_{\max}B)$ is due to the calendar rather than to the number of reports? On the very instances of Theorem 22 the answer is sharp: the same number of reports costs $T - 2K_{\max}$ when their times are fixed and only $K_{\max}$ when the defender picks them.

Proposition 24 (choosing the revelation times is exponentially cheaper). *Consider the variant of Problem 15 in which the defender has a budget of N queries rather than a schedule: at the end of any round it may spend one and receive D_ν as in Equation (34). Let the instance be played in G_3 with types whose threshold intervals form a chain under inclusion — as every instance of Theorem 22 does. Then the defender that plays the midpoint of the smallest revealed interval, and queries at the end of every round in which target 1 or 3 was attacked, spends at most K' queries and suffers $\text{Regret}(T) \leq K' \leq K_{\max}$, for every horizon and every order of appearance. By contrast Theorem 22 with $B = T/K_{\max}$ says that $N = K_{\max}$ revelations on a fixed schedule leave regret at least $T - 2K_{\max}$.*

Proof. Let D be the revealed set and $J(D)$ the smallest of their intervals, which exists because they form a chain; let μ be its midpoint and $\mathbf{p} = (\mu, 0, 1 - \mu)$. For each $\alpha_{r_1, r_2} \in D$ we have $J(D) \subseteq (r_1, r_2]$, so $r_1 < \mu < r_2$ and Equation (25) makes $\mathbf{p}$ hit it. So a round in which target 1 or 3 is attacked — the only rounds the defender queries, and by Equation (21) the only ones that cost anything — has an attacker outside D ; the query then returns a set containing it, enlarging D . As $|D| \leq K'$ throughout, there are at most K' such rounds, bounding both misses and queries. The benchmark is 0, since the midpoint of the smallest interval of all K' types hits every round and $U_d \leq 0$ everywhere; apply Equation (31). $\square$

Remark 25 (why this does not extend for free). The proposition rests on a feature of G_3 that a general SSG lacks: there, a missed round is a *certificate* that the catalogue is stale, so the defender knows exactly when to query. In general a type outside the catalogue may best-respond at $\mathbf{p}_t$ exactly as predicted and still differ elsewhere in $\mathcal{P}$, corrupting the estimates without announcing itself; and conversely a round can look anomalous merely because Algorithm 2 is exploring. Whether N self-timed queries beat the $\Theta(K_{\max}T/N)$ of Theorem 22 in a general repeated SSG is open. What Proposition 24 settles is that the lower bound is a statement about schedules, not about report counts.

11 Experiments

The experiments of Section 7 ran at $n = 3$ targets, limited not by the online algorithm but by the expert set: Algorithm 1 needs $\mathcal{E}(C; \epsilon)$ explicitly, and building it by enumerating $(n-1)$ -subsets of the defining hyperplanes costs 7.3×10^{14} linear solves at $n = 10$. Neither algorithm actually needs that set, however — each only needs to *optimize a linear function over it*, which one call to a Bayesian-Stackelberg solver does. Appendix C gives the construction and Appendix D the resulting runs at

$n = 10$ and $n = 20$; here we report what they show and then test the lower bounds directly.

At $n = 10$ and $n = 20$ with $K_{\max} = 4$, Algorithm 1 stays below the $C\sqrt{t}$ envelope of Theorem 6 throughout, and Algorithm 2 below the $Ct^{2/3}$ envelope of Theorem 18. Fitting the growth over $T \in \{250, \dots, 2000\}$ gives exponent 0.664 for block revelation — the $2/3$ of Theorem 18, to the resolution of the experiment — and 0.317 for full information, comfortably inside the $1/2$ of Theorem 6. Sweeping B at fixed T gives regret growing linearly in B with slope 0.47, well below the worst-case $2K' = 8$, which is expected: the worst case is attained only by the nested adversary of Theorem 22, exhibited separately in Figure 3(c).

11.1 The lower bounds, in simulation

Figure 3 plays the gadget G_3 of Section 9.2 for real: attacker types are built from the utility vectors of Equation (22), best responses are computed from those utilities, and the defender's payoff is read off the attacked target. The defender is the optimal one for this game, namely the midpoint version-space search of Remark 13, and the expectation over the hidden type is computed exactly, by averaging over all M candidates rather than by sampling.

Panel (a) is Theorem 12: with $M = 2^T$ the measured expected regret equals $T - 2 + (T + 2)2^{-T}$ for every $T \leq 14$, to machine precision, confirming both the lower bound and the claim in Remark 13 that it is exactly attained.

Panel (b) is Proposition 14, and it is the one panel that shows a T -dependence rather than a saturation. With $|K| = 2$ on the two-element universe, follow the leader has measured regret 1.57, 6.35 and 25.60 at $T = 16, 256$ and 4096, against the exact optimum $\frac{1}{2}\mathbb{E}|N_1 - N_2| = 1.571, 6.377, 25.531$ and the floor $\frac{1}{2}\sqrt{[T/2]} = 1.41, 5.66, 22.63$. The dotted line is the same universe with $|K| = 1$, where the regret is 0.5 at every horizon. The two curves are the phase transition of Section 9.5.

Panel (c) is Theorem 22: on the nested-interval instance the measured regret tracks $|K|(B - 2)$ across $|K| \in \{1, 2, 3, 4\}$ and $B \in \{4, 8, 16, 32\}$, so the additive $\Theta(K_{\max}B)$ of Theorem 18 is real and not an artefact of the analysis.

Panel (d) is Proposition 24, on the same instances as panel (c) with $|K| = 4$ and $T = |K|B$, so that the two schemes are compared at the same report budget $N = |K| = 4$. Scheduling the four reports on a fixed grid costs $|K|(B - 2)$, which is 120 rounds at $B = 32$; letting the defender choose their times costs exactly 4, and the measured query count is 4.00, so the defender does not even exceed its budget. The measured adaptive regret is 3.72, 3.98, 4.00, 4.00 for $B = 4, 8, 16, 32$ — it is below $|K|$ only because a lucky initial play occasionally hits the first type outright.

12 Discussion and Open Problems

Sections 9 and 10 locate the boundary of learnability for the unknown- K repeated security game. On one side, Theorem 6: an unknown attacker set of bounded size costs only $\sqrt{K_{\max}}$, provided the defender is told after each round which type attacked. On the other, Theorem 12: told nothing but the attacked target, every algorithm loses all but two of the T

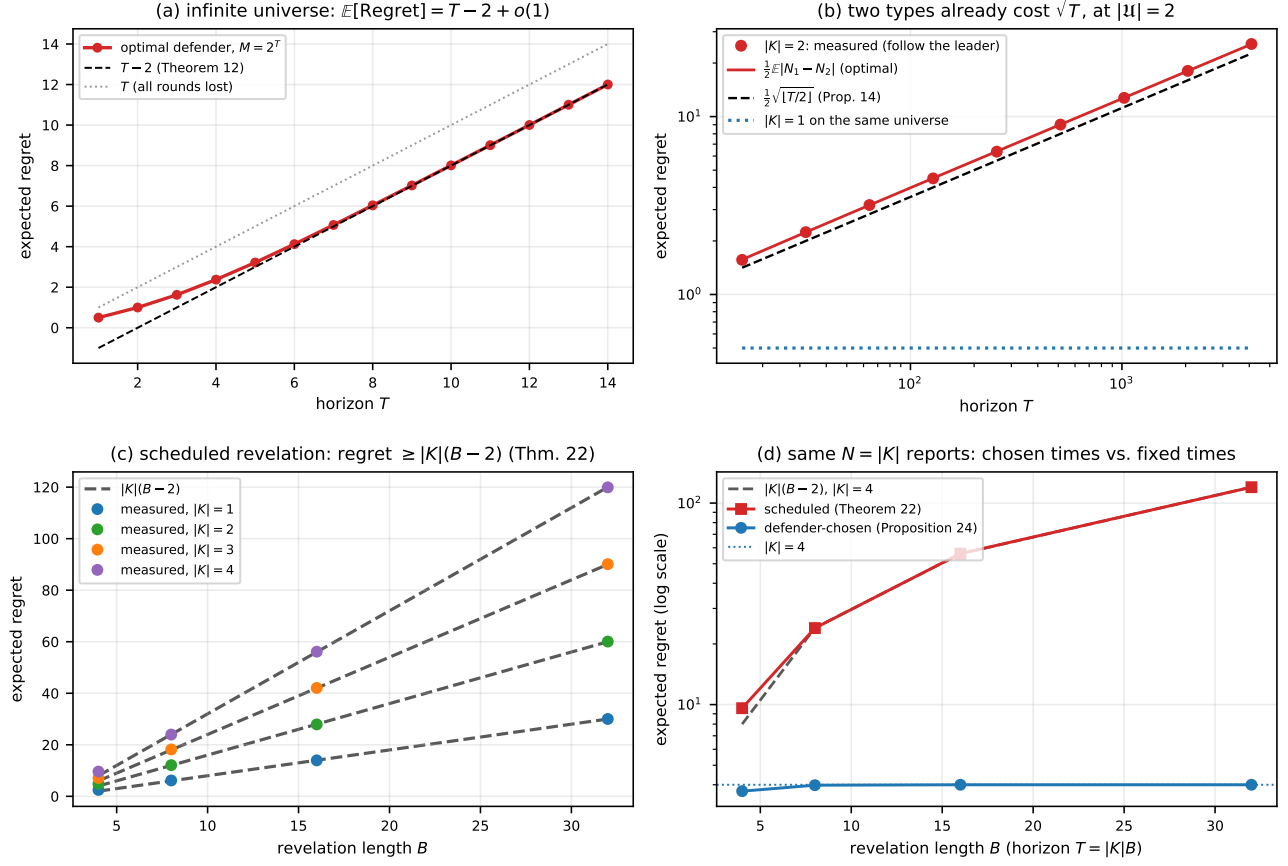


Figure 3: The gadget G_3 in simulation, against the optimal defender. (a) Theorem 12: with $|\mathcal{U}_M| = 2^T$ the regret is $T - 2 + o(1)$. (b) Proposition 14: two types on a two-element universe cost $\Theta(\sqrt{T})$, one type costs 0.5 at every horizon. (c) Theorem 22: on the nested instance the regret matches $|K|(B - 2)$ (dashed lines). (d) Proposition 24: the same $N = |K|$ reports cost $|K|(B - 2)$ on a fixed schedule and $|K|$ when the defender picks their times. Panels (b) and (d) have logarithmic vertical axes.

rounds — not because the adversary has many types, but because the defender cannot name the one it faces. Table 2 places this among the known results.

Between the two lies Problem 15, where an oracle periodically reports the *set* of types met so far — a list of names, nothing about when or how often. Theorem 18 shows this suffices: Algorithm 2 attains $O(nK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max}) + K_{\max}B)$, the partial-information rate of Balcan et al. [2, Theorem 6.1] degraded by $K_{\max}^{1/3}$ for the unknown catalogue and by an additive $K_{\max}B$ for the batching, the latter tight by Theorem 22. Corollary 21 makes this a deployment statement: the report needs to arrive only $\tilde{O}(T^{1/3})$ times over a horizon of length T — and, by the matching bound there, not fewer.

What Theorem 22 isolates is a *latency* cost: even an oracle that could say anything at all still costs $\Theta(K_{\max}B)$ when it may speak only every B rounds, because the defender plays blind between a type's arrival and the report naming it. Proposition 24 shows that this is a statement about the *schedule* and not about the number of reports — in G_3 , letting the defender choose *when* the same $K_{\max}$ reports arrive collapses the cost from $T - 2K_{\max}$ to $K_{\max}$. This places the additive term alongside the literature on online learning under delayed

feedback [10], with the difference that here the delay applies not to the loss but to the *identity* of the opponent, and that Theorem 12 makes the undelayed identity indispensable rather than merely helpful.

Four questions are left open.

Is the exponent $2/3$ necessary here? The exponent in Theorem 18 is inherited from Balcan et al. [2, Theorem 6.1] and untouched by the unknown catalogue, so improving it would improve their result. Whether $T^{2/3}$ is optimal for partial-information feedback even with K known is not known; the natural attack replaces their explore-then-exploit estimator by importance-weighted estimates on the exploit rounds, which needs a version of Lemma 17(ii) for a continuously updated estimator.

Is the factor $K_{\max}^{1/3}$ necessary? It comes from Hölder over the epochs, exactly as the $\sqrt{K_{\max}}$ of Theorem 6 comes from Cauchy–Schwarz. Theorem 22 bounds only the additive term and Proposition 14 only the T -dependence; a bound $\Omega(K_{\max}^c T^{2/3})$ matching the multiplicative price is open under both feedback models. The obvious attempt — $K_{\max}/2$ dis-

Feedback	Set K	Regret
full	known, $ K = k$	$O(\sqrt{Tn^2 k \log nk})$
partial	known, $ K = k$	$O(T^{2/3}nk \log^{1/3}nk)$
full	known, $ K = 2^{T+1} - 2$	$\Omega(T)$
full	unknown, $\leq K_{\max}$	$O(\sqrt{TK_{\max}\Lambda_n})$
partial	unknown, $\leq K_{\max}$	$\Omega(T)$, even $K_{\max}=1$
full	known, $ K = \mathcal{U} =2$	$\Theta(\sqrt{T})$
partial	unknown, $\leq K_{\max}$, block revelation	$\tilde{O}(nK_{\max}^{4/3}T^{2/3} + K_{\max}B)$

Table 2: The landscape, with $\Lambda_n = n^2 + K_{\max} \log n$. Rows 1–3 are Theorems 5.1, 6.1 and 7.1 of Balcan et al. [2]; row 4 is Theorem 6; row 5 is Theorem 12; row 6 is Proposition 14 with Theorem 6 at $k = 2$; row 7 is Theorem 18. Rows 3 and 5 are incomparable: the construction of row 3 plays T distinct types, so it needs $K_{\max} \geq T$ and is consistent with Theorem 6, whereas row 5 uses $|K| = 1$ but needs partial information and an infinite universe. The $\Omega(\sqrt{T})$ half of row 6 holds under partial information too, but the matching upper bound there is only the $O(T^{2/3})$ of row 2; closing that gap is an open problem of Section 12.

joint copies of the two-type construction of Proposition 14, one per epoch — fails instructively. Per-epoch lower bounds do not add, because the benchmark Equation (3) is a *single* fixed coverage vector: with all intervals disjoint, Lemma 10 says no $\mathbf{p}$ hits more than one type, so the global benchmark is far weaker than the sum of the per-epoch ones and the global regret correspondingly smaller. A matching lower bound must let the epochs share a good comparator, as the nested family of Theorem 22 does, while still forcing fresh learning in each.

The complementary attack is on the upper bound: avoid the restart and both factors go with it, since without epochs there is neither a Cauchy–Schwarz nor a Hölder step. Algorithms tracking a *growing* expert set [16] do that, and Section 2.2 already notes them as a drop-in for the Polynomial-Weights sub-routine. But replacing the sub-routine is not enough: Steps 4 and 5 of Section 6 also use the epoch structure to pass from the epoch-local optimum to the global $\mathbf{p}^*$. Under partial information it is worse still — the estimator of Lemma 27(ii) is defined relative to a fixed catalogue, its spanner and ambient set both changing when C grows, so estimates from before and after a discovery block are not estimates of comparable quantities.

Confounded search: $K_{\max} \geq 2$ on a finite universe. For $K_{\max} = 1$ the picture is complete: on a resolution- M universe a miss halves the version space unambiguously, so Remark 13 gives an optimum of $\Theta(\min\{T, \log_2 M\})$. For $K_{\max} \geq 2$ the defender does not know which opponent produced the observation, so the same search is corrupted. Proposition 14 settles one half — the answer must contain $\Omega(\sqrt{T})$, already at $M = 2$ — and the two upper bounds we can write down, $O(\sqrt{TK_{\max}(n^2 + K_{\max} \log n)})$ and $O(T^{2/3}nM \log^{1/3}(nM))$, contain no $\log M$ while the lower bound contains no M at all. The joint dependence is open.

Self-timed reports in a general game. Proposition 24 shows that in G_3 a defender choosing its own report times pays

$K_{\max}$ rather than $K_{\max}B$, because there a missed round certifies a stale catalogue; Remark 25 explains why that certificate is special to the gadget. Whether another trigger — a statistical test on the estimator, say — recovers a comparable saving in a general repeated SSG, and whether the $\Theta(K_{\max}T/N)$ of Corollary 21 is the truth for self-timed queries, we leave open. This is the question with the most direct operational content: it asks whether to subscribe to a periodic feed or to keep a budget of on-demand lookups.

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A The random-walk bound of Proposition 14

Let $S_T = \sum_{t \leq T} \varepsilon_t$ with $\varepsilon_t = \pm 1$ independent and uniform. We claim $\mathbb{E}|S_T| \geq \sqrt{[T/2]}$. Since $|S_m|$ is a submartingale, $\mathbb{E}|S_T| \geq \mathbb{E}|S_{2\nu}|$ for $\nu = \lfloor T/2 \rfloor$, and the classical identity $\mathbb{E}|S_{2\nu}| = 2\nu \binom{2\nu}{\nu} 4^{-\nu}$ reduces the claim to

$$\binom{2\nu}{\nu} 4^{-\nu} \geq \frac{1}{2\sqrt{\nu}}. \quad (50)$$

This holds with equality at $\nu = 1$ and follows by induction: the ratio of consecutive left-hand sides is $(2\nu + 1)/(2\nu + 2)$ and of

right-hand sides $\sqrt{\nu/(\nu + 1)}$, and $(2\nu + 1)^2(\nu + 1) \geq \nu(2\nu + 2)^2$ reduces to $\nu + 1 \geq 0$. Hence $\mathbb{E}|S_T| \geq 2\nu/(2\sqrt{\nu}) = \sqrt{\nu}$.

B An anytime partial-information learner

One property of $\mathcal{A}(C)$ that the proof of Theorem 6 obtained for free has to be arranged explicitly here. There the inner learner was Polynomial Weights, which by Proposition 3 is *anytime*, so the fact that an epoch’s length is revealed only when the epoch ends cost nothing. The algorithm of Balcan et al. [2, Theorem 6.1], by contrast, tunes its number of estimation blocks to a horizon supplied in advance, and the epochs of Definition 16 again have lengths that the defender does not know when they begin. The following lemma supplies the anytime form that Theorem 18 will use. Its proof takes Balcan et al. [2, Theorem 6.1] as a black box except for one structural feature of the proof of their Lemma 6.2, which we quote rather than repeat.

Lemma 26 (an anytime partial-information learner). *Let C be a finite catalogue, let T be the horizon of Problem 15, and let $\epsilon \leq 1/T$. There is an algorithm $\mathcal{A}(C)$ for the partial-information setting with known attacker set C , running over $\mathcal{E}(C; \epsilon)$ and not receiving its own horizon in advance, and an absolute constant c_4 (the constants c_1, c_2, c_3 are already spoken for in Section 6), such that for every oblivious sequence $a'_1, a'_2, \dots$ with $a'_t \in C$ for all t and for every $1 \leq L \leq T$ simultaneously,*

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[\sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_4 L^{2/3} n |C| \log^{1/3}(n|C|). \end{aligned} \quad (51)$$

Proof. (a) The bound of Balcan et al. [2, Theorem 6.1] already holds at every prefix of a run tuned to a horizon H . Their algorithm divides $[1, H]$ into $Z = n(H^2 \log(n|C|))^{1/3}$ estimation blocks of H/Z rounds each, with Z a function of H , n and $|C|$ only. Dispose first of the degenerate range $Z \geq H$, in which there are not enough rounds to form the blocks at all: that range is $H \leq n^3 \log(n|C|)$, and there the trivial bound of two per round gives regret at most $2H = 2H^{1/3} \cdot H^{2/3} \leq 2n \log^{1/3}(n|C|) H^{2/3}$, which is below the claim for any $c_0 \geq 2$. So assume $Z \leq H$. The chain of inequalities in their proof of Lemma 6.2 bounds the algorithm’s loss over the first τ_0 blocks by the loss of the best fixed action over those same blocks plus

$$\frac{H}{Z} R_{\tau_0, \kappa} + \tau_0 |C|,$$

where $\kappa = n|C|$ is the range of the loss estimator [2, Lemma 6.4], $R_{\tau_0, \kappa}$ is the regret of the full-information sub-routine after τ_0 estimated loss vectors, and the term $\tau_0 |C|$ pays for the $|C|$ exploration rounds of each block. Both summands are non-decreasing in τ_0 and are maximized at $\tau_0 = Z$; and the sub-routine of Proposition 3 is anytime, so $R_{\tau_0, \kappa} = O(\sqrt{\tau_0 \kappa \log |\mathcal{E}(C; \epsilon)|})$ holds for every τ_0 and not only for $\tau_0 = Z$. Reading the chain at an arbitrary $\tau_0 \leq Z$ therefore reproduces their bound with H in place of T . A prefix that stops inside a block contributes at most two per round for the at most H/Z rounds of that block, and $H/Z \leq H^{2/3} n|C| \log^{1/3}(n|C|)$,

so this is absorbed. The discretization error incurred by optimizing over $\mathcal{E}(C; \epsilon)$ instead of $\mathcal{P}$ is $2\epsilon L \leq 2$ by Lemma 1 and $\epsilon \leq 1/T$, and is absorbed as well; note that ϵ enters the size bound of Lemma 2 not at all, so taking it smaller than Balcan et al. [2] do is free. Hence there is an absolute constant c_0 such that a run tuned to H satisfies Equation (51) with $L^{2/3}$ replaced by $H^{2/3}$, simultaneously for every $L \leq H$.

(b) *Doubling.* Write $\phi = n|C| \log^{1/3}(n|C|)$ for the catalogue-dependent factor in Equation (51), and let $\mathcal{A}(C)$ run in stages $s = 0, 1, 2, \dots$, where stage s occupies 2^s consecutive rounds and is a fresh instance tuned to the horizon $H_s = 2^s$. Fix L and let S be the stage containing round L . Stages $0, \dots, S-1$ occupy $2^S - 1$ rounds, all of them at most L , so $2^S \leq L$. For any $\mathbf{p} \in \mathcal{P}$ the sum over $[1, L]$ splits across stages, so the maximum over $\mathcal{P}$ of the sum is at most the sum over stages of the per-stage maxima; subtracting the algorithm's payoff stage by stage gives

$$R_{[1, L]} \leq \sum_{s=0}^S R^{(s)},$$

where $R^{(s)}$ denotes the regret of the instance of stage s over that part of stage s which lies in $[1, L]$. Each $R^{(s)}$ is bounded by (a) with $H = 2^s$ — the restriction of an oblivious sequence to a stage is oblivious and still takes values in C — so

$$R_{[1, L]} \leq c_0 \phi \sum_{s=0}^S 2^{2s/3} \leq \frac{2^{2/3}}{2^{2/3} - 1} c_0 \phi 2^{2S/3},$$

and $2^S \leq L$ gives Equation (51) with $c_4 = 2^{2/3} c_0 / (2^{2/3} - 1) < 2.71 c_0$. $\square$

One caveat about the factor $n|C|$ in Equation (51) should be recorded, since Lemma 26 is the one place in this report where the proof of Balcan et al. [2, Theorem 6.1] is reopened rather than quoted. That factor is inherited, through their Lemma 6.2, from the form $R_{L', \kappa} \leq \sqrt{L' \kappa \log |\mathcal{M}|}$ in which Proposition 3 states the full-information guarantee for losses of range κ . The textbook guarantee for Polynomial Weights on losses in $[-\kappa, \kappa]$ is instead $O(\kappa \sqrt{L' \log |\mathcal{M}|})$, obtained by rescaling to $[0, 1]$, and the two agree only at $\kappa = 1$. In Theorem 6 the loss range is indeed $\kappa = 1$ and nothing is at stake; here it is $\kappa = n|C|$ [2, Lemma 6.4], and the difference matters. We state Lemma 26 and Theorem 18 with the factor exactly as Balcan et al. [2, Theorem 6.1] delivers it. A reader who prefers the rescaling bound should re-optimize their Lemma 6.2 with $R_{L', \kappa} = \kappa \sqrt{L' \log |\mathcal{M}|}$, which gives $O(L'^{2/3} |S|^{1/3} \kappa^{2/3} \log^{1/3} |\mathcal{M}|)$ and hence $n^{4/3} |C|^{4/3}$ in place of $n|C|$ in Equation (51), and $O(n^{4/3} K'^{5/3} T^{2/3} \log^{1/3}(nK') + 2K'B)$ in place of Equation (36). Nothing else in this section changes: the epoch decomposition, the Hölder step that produces the $K'^{1/3}$, and the additive $2K'B$ are all independent of which form is used.

C An oracle-efficient implementation

For a catalogue C and $\mathbf{p} \in \mathcal{P}$ define the *lift*

$$y_{\alpha, i}(\mathbf{p}) = \mathbb{I}\{b_{\alpha}(\mathbf{p}) = i\}, \quad w_{\alpha, i}(\mathbf{p}) = y_{\alpha, i}(\mathbf{p}) p_i \quad (52)$$

for $(\alpha, i) \in C \times N$, and let $\mathcal{S}(C) = \{(y(\mathbf{p}), w(\mathbf{p})) : \mathbf{p} \in \mathcal{P}\} \subseteq \mathbb{R}^{2|C|n}$.

Lemma 27 (both algorithms are linear over the lift). *Assume the types in C have pairwise distinct best-response maps, so that the set $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$ of Balcan et al. [2, Section 6.3] has rank $|C|$ and its barycentric spanner Λ is invertible. Let $\omega \in \mathbb{R}^C$ be any vector of weights on the catalogue. Then*

$$\sum_{\alpha \in C} \omega_{\alpha} U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}) = \sum_{\alpha \in C} \sum_{i \in N} \left[\omega_{\alpha} u_d^u(i) y_{\alpha, i}(\mathbf{p}) + \omega_{\alpha} (u_d^c(i) - u_d^u(i)) w_{\alpha, i}(\mathbf{p}) \right], \quad (53)$$

which is a linear function of the lift. Moreover:

1. the loss used by Algorithm 1 at round t is $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$, i.e. Equation (53) with $\omega = -\mathbb{I}_{\{a_t\}}$;
2. the estimator $\hat{c}_{\tau}$ of Balcan et al. [2, Eq. 2 and Lemma 6.4], used inside Algorithm 2, is Equation (53) with $\omega = g_{\tau}$ — and the loss it induces is its negation, i.e. $\omega = -g_{\tau}$ — where $g_{\tau} = \Lambda^{-1} \hat{p}_{\tau}$, Λ is the matrix whose rows are the barycentric spanner $b_1, \dots, b_{|C|}$ of Balcan et al. [2, Section 6.3] and $\hat{p}_{\tau} \in \{0, 1\}^{|C|}$ collects that block's exploration outcomes. The vector g_{τ} is an unbiased estimator of the per-round type frequencies in the block.

Proof. For each α exactly one of $y_{\alpha, 1}, \dots, y_{\alpha, n}$ equals 1, so $U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}) = \sum_i y_{\alpha, i}(\mathbf{p}) U_d(i, \mathbf{p})$. By Equation (1),

$$\begin{aligned} y_{\alpha, i} U_d(i, \mathbf{p}) &= u_d^c(i) y_{\alpha, i} p_i + u_d^u(i) y_{\alpha, i} (1 - p_i) \\ &= u_d^u(i) y_{\alpha, i} + (u_d^c(i) - u_d^u(i)) w_{\alpha, i}, \end{aligned}$$

and summing against ω gives Equation (53). Item (i) is the special case $\omega = -\mathbb{I}_{\{a_t\}}$.

For (ii), recall that Balcan et al. [2, Eq. 2] sets $\hat{c}_{\tau}(\mathbf{p}) = \sum_i \sum_j \lambda_j(\mathbb{I}_{(\sigma=i)}) \hat{p}_{\tau}(b_j) U_d(i, \mathbf{p})$, where σ is the response profile of $\mathbf{p}$ and $\lambda(w)$ is the (unique, since Λ is invertible) vector of coefficients with $\sum_j \lambda_j(w) b_j = w$, i.e. $\lambda(w)^{\top} \Lambda = w^{\top}$. Hence $\sum_j \lambda_j(w) \hat{p}_{\tau}(b_j) = \lambda(w) \cdot \hat{p}_{\tau} = w^{\top} \Lambda^{-1} \hat{p}_{\tau} = w \cdot g_{\tau}$. The α -th coordinate of $\mathbb{I}_{(\sigma=i)}$ is exactly $y_{\alpha, i}(\mathbf{p})$, so

$$\begin{aligned} \hat{c}_{\tau}(\mathbf{p}) &= \sum_i \left(\sum_{\alpha} g_{\tau, \alpha} y_{\alpha, i}(\mathbf{p}) \right) U_d(i, \mathbf{p}) \\ &= \sum_{\alpha} g_{\tau, \alpha} U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}), \end{aligned}$$

which is Equation (53) with $\omega = g_{\tau}$. Finally, Balcan et al. [2, Lemma 6.3] gives $\mathbb{E}[\hat{p}_{\tau}(b_j)] = f_{\tau}(b_j)/|B_{\tau}|$ and $f_{\tau}(w) = w \cdot m_{\tau}$ with m_{τ} the vector of type counts in the block, so $\mathbb{E}[\hat{p}_{\tau}] = \Lambda m_{\tau}/|B_{\tau}|$ and $\mathbb{E}[g_{\tau}] = m_{\tau}/|B_{\tau}|$. $\square$

Lemma 27 says that both algorithms are instances of online linear optimization over $\mathcal{S}(C)$, whose single primitive is

$$\text{ORACLE}(A, \Delta) = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{\alpha, i} (A_{\alpha, i} y_{\alpha, i}(\mathbf{p}) + \Delta_{\alpha, i} w_{\alpha, i}(\mathbf{p})). \quad (54)$$

We therefore replace the Polynomial-Weights sub-routine of Proposition 3 by Follow the Perturbed Leader over $\mathcal{S}(C)$, which calls Equation (54) once per round — and no times at

all on the exploration rounds of Algorithm 2, which play a fixed spanner strategy. This is a change of implementation and not of algorithm: Balcan et al. [2, Section 3.1] explicitly allow any black box satisfying their Proposition 1, and name Follow the Lazy Leader of Kalai and Vempala [11] as an admissible choice. We run it with the perturbation schedule of Kalai and Vempala [11], in its anytime form because the segment lengths are not known in advance. The three problem constants that schedule needs take, on $\mathcal{S}(C)$, the values $\|Y - Y'\|_1 \leq 4|C|$ for the ℓ_1 -diameter of the decision set, $|Y \cdot s| \leq 1$ for the per-round loss, and $\|s\|_1 \leq 3n$ for the ℓ_1 -norm of a one-round loss vector, the last because $\sum_i (|u_d^u(i)| + |u_d^c(i) - u_d^u(i)|) \leq 3n$.

Grouping Equation (54) by the response profile $\sigma : C \rightarrow N$ makes the objective linear on each cell $\mathcal{P}_\sigma(C)$ of Equation (6), so Equation (54) is a maximum of $n^{|C|}$ linear programs in n variables — polynomial in n for a fixed catalogue, and the natural extension to a set of types of the multiple-LP method of Conitzer and Sandholm [5]. In the experiments we instead solve one mixed-integer program with $n|C|$ binaries in the style of the Bayesian-Stackelberg MILP of Paruchuri et al. [17], using the profile enumeration as a correctness reference. We do not claim polynomial time in $|C|$: the Bayesian Stackelberg problem is NP-hard [5]. The claim is polynomial time in n , which is the parameter that made $\mathcal{E}(C; \epsilon)$ unusable. Both back-ends solve over cells shrunk by a strictness margin $\gamma = 10^{-6}$, so that the returned $\mathbf{p}$ realizes its profile under any tie-breaking rule; this plays the role of the ϵ -approximation in Lemma 1. The rank hypothesis of Lemma 27 is not an extra assumption on the game but a normalisation of the catalogue: two types with the same best-response map on all of $\mathcal{P}$ are indistinguishable to the defender’s payoff, to the benchmark Equation (3) and to the feedback, so Algorithm 2 keeps one representative of each best-response map in C and discards the duplicates. This can only shrink the catalogue, so $|C_j| \leq K'$ and every bound of Section 10 is unaffected; without it $\mathcal{W}$ can have rank below $|C|$ and Λ^{-1} in Lemma 27(ii) need not exist. The implementation does not de-duplicate but detects the failure: if the sampled pool admits no full-rank spanner it records a rank failure for that catalogue and skips exploration until the catalogue next changes. Over the runs behind Figure 5 no rank failure occurred, which is what one expects of independently drawn utility vectors. As in Section 7, the experiments take $\mathcal{P} = \{\mathbf{p} \geq 0 : \sum_i p_i = 1\}$ with one resource; the algorithm and the benchmark $\mathbf{p}^*$ optimize over the same set, so the regret they define is the one of Equation (4). Every non-asymptotic claim of Sections 9 and 10, together with the oracle and estimator claims of this subsection, is re-checked by `code/verify_theory.py`, which prints a pass or fail for each; the measured quantities of Appendices C.1, D.1 and D.2 and Section 11.1 — fitted exponents, envelope constants and timings — are printed by the plotting and benchmarking scripts that produce the corresponding figure or table, and are reproduced by rerunning those.

One further gap between Algorithm 2 and its implementation should be stated. The set $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$ of Balcan et al. [2, Section 6.3] has one element per pair (cell, target) and we do not enumerate it; instead we sample coverage vectors, read off the vectors $\mathbb{I}_{(\sigma=i)}$ they realize together with their witnesses $(\mathbf{p}_b, i_b)$, and run the greedy determinant maximization of Awerbuch and Kleinberg [1] on that pool. The consequence

n	$ C $	$\binom{H}{n-1}$	build $\mathcal{E}$	$n^{ C }$ LPs	MILP
3	4	1.1×10^2	6 ms ($ \mathcal{E} = 61$)	32 ms	16 ms
5	4	1.5×10^5	307 s ($ \mathcal{E} = 13118$)	230 ms	64 ms
10	4	7.3×10^{14}	infeasible	3.1 s	133 ms
20	4	5.9×10^{37}	infeasible	61 s	0.88 s
30	4	1.4×10^{63}	infeasible	infeasible	0.89 s
40	4	1.2×10^{90}	infeasible	infeasible	1.56 s

Table 3: Cost of one step, with $H = H(n, |C|) = n + |C| \binom{n}{2}$. Column 3 is the number of $(n-1)$ -subsets that the exhaustive construction of $\mathcal{E}(C; \epsilon)$ must solve. Columns 5 and 6 are the two back-ends for Equation (54); wherever both ran they returned the same optimum to 2×10^{-6} . Median of 10 calls on an otherwise idle machine; `code/bench_oracle.py`. The MILP is not monotone in n because it is an exact branch-and-bound solve, not a fixed amount of work.

is confined to the *range* of the estimator: a spanner of a sub-pool need not give coefficients $\lambda \in [-1, 1]$ for every $w \in \mathcal{W}$, so the bound $\hat{c}_\tau \in [-n|C|, n|C|]$ of Balcan et al. [2, Lemma 6.4] is not guaranteed by the implementation. Unbiasedness — Lemma 27(ii), which is what the estimator’s correctness rests on — needs only that Λ be invertible, and holds exactly; it is the check reported in the previous paragraph. For the same reason the implementation does not scale the Follow-the-Perturbed-Leader perturbation by that worst-case range: inside Algorithm 2 it uses the schedule $\epsilon_t = \sqrt{4|C|/(3nt)}$ listed above, which is what the range of the estimates we actually compute warrants, rather than the range $n|C|$ that Balcan et al. [2, Lemma 6.4] proves in the worst case. This is a tuning choice and not a claim: Theorem 18 rests on Lemma 26 and on their Lemma 6.4, not on the constants used here, and what Figure 5 sets out to test is whether the predicted *exponent* appears in a run. Over the 60 random instances with $n \in \{3, 4, 5\}$ and $|C| \in \{1, 2, 3\}$ that `verify_theory.py` runs, the two back-ends and a brute-force maximization over the enumerated $\mathcal{E}(C; \epsilon)$ agree to within 1.4×10^{-6} , and a longer run of 150 instances does not push that figure up. We also checked Lemma 27(ii) directly: over the 4×10^4 replications of a single estimation block at $n = 6$, $|C| = 3$ that the same script runs, the mean of $|B_\tau| g_\tau$ reproduced the true type counts of the block to within 0.05 out of 40, comfortably inside the three-standard-error band of 1.1 that Monte-Carlo noise allows; a separate run at 2×10^5 replications gave a largest error of 0.09 against a standard error of 0.16, which is what an unbiased estimator should do.

C.1 Tractability

Table 3 times the three routines. Building $\mathcal{E}(C; \epsilon)$ already takes 307 seconds at $n = 5$, $|C| = 4$ and is out of reach beyond that, whereas a single oracle call costs 133 ms at $n = 10$, 0.9 s at $n = 20$ and 1.6 s at $n = 40$. Since neither algorithm makes more than one oracle call per round — and Algorithm 2 makes none at all on its exploration rounds — this is what moves the experiments from $n = 3$ to $n = 20$.

D Experiments at $n = 10$ and $n = 20$

D.1 Full information at $n = 10$ and $n = 20$

Figure 4 reruns the two experiments of Section 7 with the oracle implementation, at $n = 10$ and $n = 20$ instead of $n = 3$, with $K_{\max} = 4$ and $T = 1500$. Panels (a) and (b) reproduce Figure 1 under the staggered-uniform adversary: the per-round regret spikes shortly after each of the four phase boundaries — a new type becomes available at a boundary and is first drawn a few rounds later, which is the discovery round — and then decays inside the epoch that follows, exactly the structure that Steps 1–6 of the proof of Theorem 6 describe. Panels (c) and (d) reproduce Figure 2 under the uniform-random adversary: the cumulative regret stays below the $C\sqrt{t}$ envelope of Theorem 6 throughout, with $C = 1.27$ at $n = 10$ and $C = 1.71$ at $n = 20$, and it is 8.0 and 11.6 times smaller than the uniform baseline.

We keep the two adversaries separate for the same reason Section 7 does, and it is worth spelling out. The staggered-uniform adversary is non-stationary by design — the type mixture changes at every phase boundary — so the best *fixed* strategy in hindsight of Equation (3) is a weak comparator, and an algorithm that tracks the current phase can and does beat it. At $n = 10$ the cumulative regret against $\mathbf{p}^*$ came out negative on one of the four seeds (-35.3), and the four-seed mean was -2.6 . That says nothing against Theorem 6, which is an upper bound, but it makes a cumulative-regret plot on that adversary uninformative. The uniform-random adversary is stationary, so $\mathbf{p}^*$ is the right comparator, and panels (c) and (d) use it. The benchmark $\mathbf{p}^*$ is itself computed with one oracle call, so even *evaluating* the regret at $n = 20$ was out of reach before Appendix C.

D.2 Block revelation at $n = 10$ and $n = 20$

Figure 5 runs Algorithm 2 under genuine partial-information feedback: the simulator passes the algorithm the attacked target only, and at the end of each revelation block calls it with the *set* of types seen so far and nothing else. The set is handed over as a list for convenience, but the algorithm reads it only through membership tests, so neither the order in which the types first attacked nor any frequency information can leak through it. Panels (a) and (b) show cumulative regret against the $Ct^{2/3}$ envelope of Theorem 18 under the staggered-uniform adversary, which is the interesting one here because it spreads the discovery blocks across the horizon and so produces the $\Theta(K_{\max})$ epochs that Step 8 of the proof pays for; the envelope constants are $C = 1.72$ at $n = 10$ and $C = 1.25$ at $n = 20$.

Panel (c) sweeps the revelation length B over six doublings at fixed T . The measured growth is linear in B , as Theorems 18 and 22 predict — a least-squares fit over $B \in [4, 256]$ has slope 0.47 — but that slope is far below the worst-case $2K' = 8$, and the panel draws both. The gap is expected rather than a discrepancy: the worst case is attained only by the nested adversary of Theorem 22, which Figure 3(d) exhibits separately, and on a random instance most of the rounds of a discovery block are not in fact wasted, because a strategy tuned to the types already in the catalogue is usually not terrible against a new one.

Panel (d) compares the two feedback models as T grows, on the stationary uniform-random adversary for the reason given in Appendix D.1. Over $T \in \{250, 500, 1000, 2000\}$ the block-revelation regret grows with fitted exponent 0.664, which is $2/3$ to within the resolution of the experiment and is exactly the exponent of Theorem 18; the full-information regret grows with exponent 0.317, comfortably inside the $1/2$ of Theorem 6. The asymmetry is not an accident of the instances. Algorithm 2 pays a floor of $|C|$ exploration rounds in every estimation block whatever the sequence does, so its $T^{2/3}$ is visible even on an easy sequence, whereas Algorithm 1 has no such floor and, as already observed in Section 7, runs well inside its worst case against a stochastic adversary.

One implementation choice is worth stating, because it is where asymptotics and simulation disagree. Balcan et al. [2, Algorithm 1] uses $Z = n(T^2 \log(n|C|))^{1/3}$ estimation blocks, i.e. blocks of $T/Z = T^{1/3}/(n \log^{1/3}(n|C|))$ rounds, which at $n = 10$, $|C| = 4$ and $T = 2000$ is less than one round — fewer, certainly, than the $|C| = 4$ exploration rounds a block must hold, since the tuning of Balcan et al. [2, Theorem 6.1] is asymptotic. We therefore use blocks of $\max\{|C| + 1, \lceil T^{1/3} \rceil\}$ rounds, which has the same order in T . At $T = 2000$ and $K_{\max} = 4$ that is 13 rounds, of which 4 are spent exploring, so roughly 31% of all rounds are exploration; this is the main reason the measured regret in panels (a) and (b) is an order of magnitude above the full-information regret of Figure 4 at the same horizon, and it is a constant-factor effect, not a change of exponent.

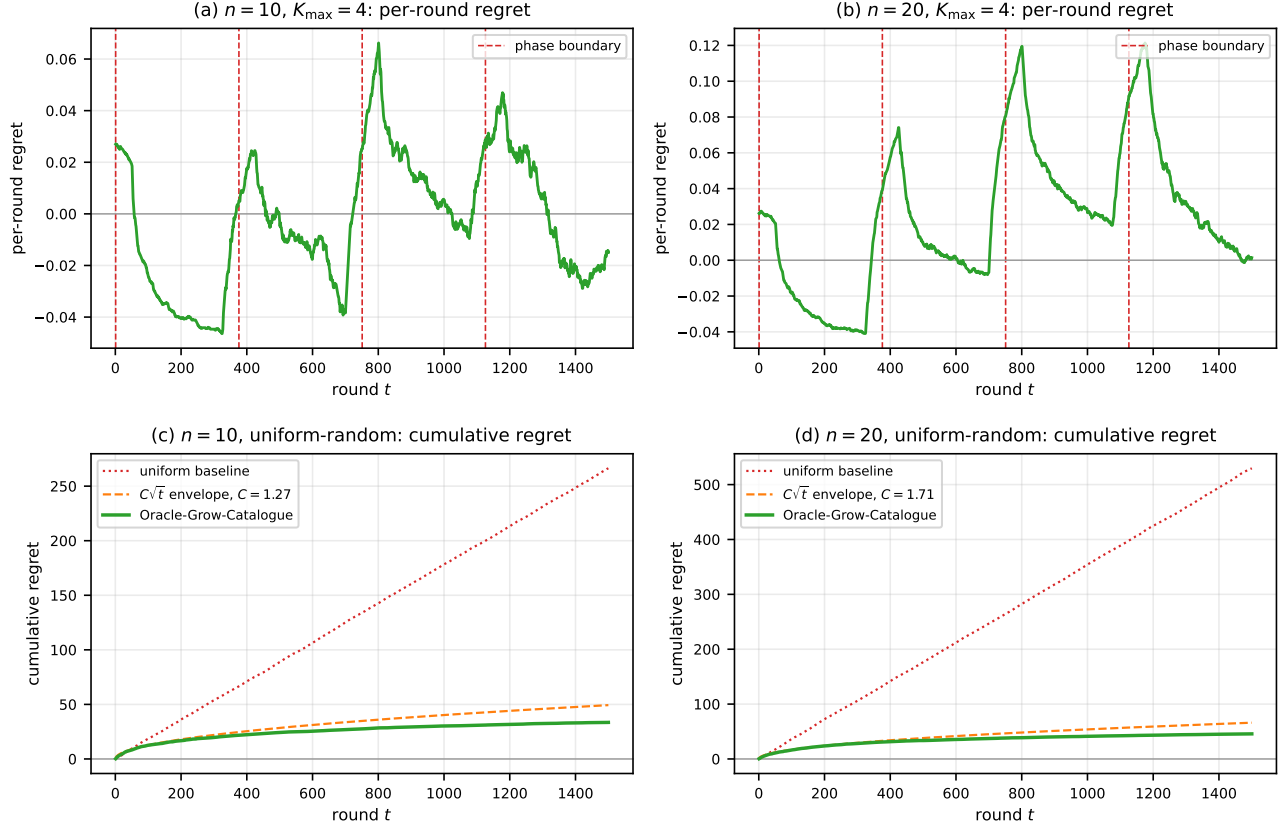


Figure 4: Algorithm 1 with the oracle implementation of Appendix C, $K_{\max} = 4$, $T = 1500$, mean over 4 seeds. (a), (b) per-round regret under the staggered-uniform adversary, rolling window 101; red dashed lines mark the phase boundaries, at which a new type becomes available. (c), (d) cumulative regret under the uniform-random adversary, against the $C\sqrt{t}$ envelope of Theorem 6 and the uniform baseline.

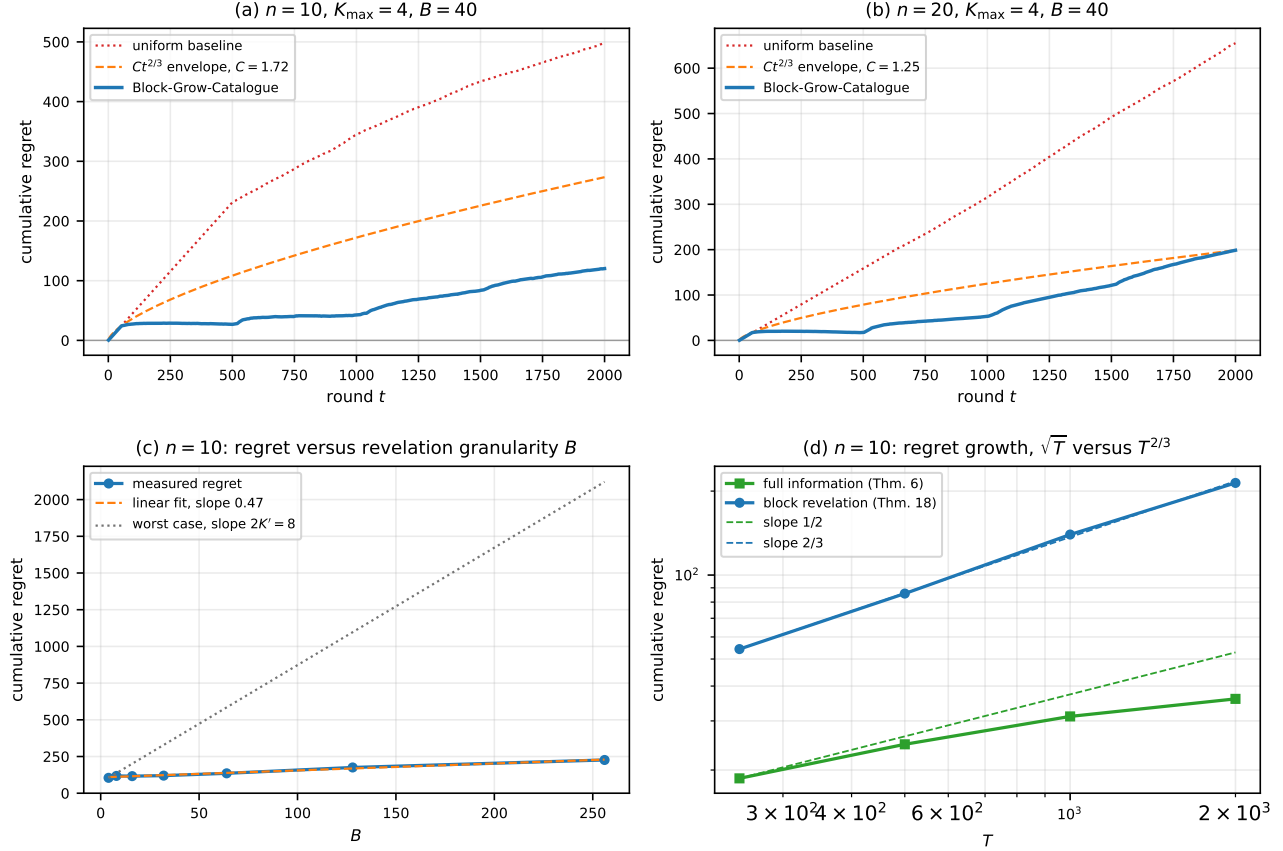


Figure 5: Algorithm 2 under partial-information feedback with block revelation, $K_{\max} = 4$, $T = 2000$, mean over 4 seeds. (a)–(c) staggered-uniform adversary; (d) uniform-random adversary. (a), (b) cumulative regret against the $Ct^{2/3}$ envelope of Theorem 18. (c) final regret as a function of the revelation length B , against a least-squares fit and against the worst-case slope $2K'$ of Theorem 18. (d) full information (Theorem 6) against block revelation (Theorem 18) as T grows; fitted exponents 0.317 and 0.664.